

Pre-Symptomatic Crop Intelligence: A Closed-Loop Framework for Anticipatory, Confidence-Aware Decision-Making in Site-Specific Crop Protection

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Pre-Symptomatic Crop Intelligence: A Closed-Loop Framework for Anticipatory, Confidence-Aware Decision-Making in Site-Specific Crop Protection

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Abstract

Purpose: Symptom-triggered crop protection acts only after damage is committed, and the intervention window has narrowed. This review reframes pre-symptomatic sensing from a detection problem into a closed decision loop, establishing the physiological lead time of a signal, set against its detection confidence and the latency of the response it can trigger, as the organizing principle for anticipatory, site-specific decision-making. **Methods:** A systematic-narrative synthesis was conducted across major bibliographic databases through June 2026. Studies reporting pre-symptomatic capability under field or realistic conditions were retained and coded onto a coupled lead-time $\times$ confidence $\times$ actionability framework spanning sensing, inference, and actuation. **Results:** Optical modalities were found to dominate the evidence base, while electrophysiological and volatile signals extended achievable lead time. Single modalities were insufficient to separate biotic from abiotic stress, motivating heterogeneous fusion. Edge inference and temporal onset forecasting remained immature, detection confidence was rarely quantified, and the sensing-to-actuation loop was seldom closed. Reported performance degraded sharply from laboratory to field, particularly in perennial and smallholder systems. **Conclusions:** A unifying Pre-Symptomatic Crop Intelligence framework is proposed, governed by the principle that system value is bounded by the weakest of lead time, detection confidence, and response latency; priorities identified include lead-time-labeled benchmarks, uncertainty-aware inference, field-robust fusion, and economic evaluation for perennial crops.

Keywords: decision support systems; edge artificial intelligence; multimodal sensor fusion; pre-symptomatic detection; precision crop protection; site-specific management

Introduction

Plant diseases are responsible for approximately USD 220 billion in global agricultural losses each year, representing an annual harvest loss of 10% to 16% that directly threatens global food security (Shafay et al., 2025; Upadhyay et al., 2025). These staggering economic and agricultural statistics, originally compiled by the Food and Agriculture Organization (FAO) and international pathology surveys, underscore the critical vulnerability of modern agricultural systems under escalating global demand and climatic volatility (Shafay et al., 2025). Despite decades of progress in agricultural technology, crop protection continues to operate on a predominantly reactive basis. Traditional management protocols rely on visual scouting or passive crop monitoring, meaning that physical

interventions—such as fungicide, herbicide, or insecticide spraying—are only initiated after macroscopic symptoms (such as chlorosis, necrosis, wilting, or localized tissue decay) have become clearly visible to human operators or standard RGB imaging sensors (W. Li et al., 2025; Upadhyay et al., 2025).

This visual-stage reliance represents a fundamental limitation in precision agriculture. By the time foliar symptoms are visible, significant cellular and physiological degradation is already committed, pathogen populations have expanded, and the biological window for highly effective, preventative, and low-dose chemical or biological control has severely narrowed. (W. Li et al., 2025; Moustaka & Moustakas, 2023). This lag in response not only compromises potential harvest yields but also forces growers to rely on intensive, broad-spectrum chemical cover sprays, which increase production costs, accelerate pesticide resistance, and lead to off-target environmental drift. (W. Li et al., 2025). To achieve a truly sustainable and predictive smart farming paradigm, crop protection must shift its focus from the rapid classification of visual damage to the anticipatory detection of physiological stress precursors.

This anticipatory frontier is defined by the "pre-symptomatic window," which represents the chronological duration between the initial physiological or biochemical perturbation of the plant and the eventual macroscopic expression of visual symptoms. (Chaerle et al., 2004; Moustaka & Moustakas, 2023).

During this sub-visual phase, the plant undergoes a sequence of defensive, metabolic, and biophysical changes: stomatal closure, transpiration rate reduction, transmembrane potential depolarization, and the down-regulation of light-harvesting Photosystem II (PSII) pathways. (Chaerle et al., 2004; Moustaka & Moustakas, 2023).

These early defense responses alter the plant's thermal and spectral reflectance signatures, as well as its vascular electrical potentials, before any change in green leaf pigments. (Chaerle et al., 2004; Tran et al., 2019).

By capturing these sub-visual signatures using advanced multi-sensor arrays—such as high-resolution chlorophyll fluorescence (ChlF) kinetics, thermal imaging, narrow-band hyperspectral imaging (HSI), and in-vivo plant electrophysiology—the crop protection system can identify stress vectors days before they manifest visibly (Moustaka & Moustakas, 2023; Zhu et al., 2017).

However, translating these fleeting physiological signals into effective crop protection remains a critical bottleneck. The vast majority of the pre-symptomatic literature is highly fragmented and isolated in laboratory or greenhouse settings. Sensing studies focus heavily on comparing the classification accuracies of different machine learning models on uniform leaf-on-plate images, treating detection as an end in itself rather than as the trigger for a physical, site-specific field action. A pre-symptomatic signal is agronomically useless unless it can be translated into a timely, confident, and localized physical intervention.

This review presents this relationship as a conceptual framework synthesized from the reviewed literature — an organizing principle for evaluating pre-symptomatic systems — rather than an empirically fitted or validated mathematical model.

This review argues that the realized agronomic and economic value (V) of a pre-symptomatic plant protection system is not determined by its most advanced sensor or model, but is strictly bounded by the weakest of three coupled cyber-physical quantities: physiological *lead time* (L_t), calibrated detection *confidence* (C_d), and actuation *actionability* (A_c). Formally:

$$V = f(L_t, C_d, A_c) \quad \text{where} \quad \lim_{\min(L_t, C_d, A_c) \rightarrow 0} V = 0$$

where L_t is the physiological lead time of the stress signal (defined as the duration between the onset of sub-visual physiological stress and the eventual manifestation of visible foliar symptoms) (Xu et al., 2026; Zhu et al., 2017), C_d is the calibrated detection confidence or classification reliability under heterogeneous field shifts (Hasan et al., 2025; Xiang et al., 2026), and A_c is the actuation actionability, defined as the inverse of the response latency (R_l) required to execute the physical intervention in the field (Li, W. et al. 2025): $A_c = \frac{1}{R_l}$

This mathematical coupling reveals that a failure in any single node of the loop completely neutralizes the utility of the entire system. For instance, if an advanced sensor detects water-deficit or fungal stress four days before visible symptoms emerge ($L_t = 4$ d) (Xu et al., 2026), but the downstream edge-inference classifier suffers an

uncalibrated accuracy collapse under open-field lighting conditions, shadow casting, and background clutter ($C_d \rightarrow 0$) (Xiang et al., 2026), the realized system value collapses to zero. Low-confidence, uncalibrated alerts trigger false-alarm cascades that growers cannot economically or operationally justify. Conversely, even if a model identifies stress with absolute confidence ($C_d \rightarrow 1$) and high lead time ($L_t = 4$ d) (Xu et al., 2026), but the mechanical delivery mechanism has a sluggish response latency ($R_t \rightarrow \infty$; $A_c \rightarrow 0$), the physical treatment will be executed too late, missing the pre-symptomatic action window and resulting in irreversible foliar damage (W. Li et al., 2025).

Achieving loop symmetry—where the physical response latency of the sprayer or irrigator is mathematically balanced with the physiological lead time of the biological signal and the calibrated confidence of the edge processor—represents the central design principle of the PSCI architecture (Kallenberg et al., 2025).

This review addresses these challenges by making four key contributions to the precision agriculture literature:

- **The Closed-Loop Cyber-Physical Reference Architecture:** We define the end-to-end sensing–decision–actuation–feedback reference architecture required to close the loop, shifting from a passive "perception–decision–execution" model to an active cyber-physical loop that links sub-visual precursors to variable-rate actuators.
- **The $L_t \times C_d$ Decision-Space Framework:** We transition the evaluation of precision agriculture tools from "which sensing hardware is better" to "what decision is mathematically and risk-justified," partitioning the physiological lead-time $\times$ calibrated confidence plane into risk-graded operational regimes (Act Now, Monitor, Defer) (Fig. 6).
- **The Predictive Field-State Feedback Node:** We close the loop by integrating a predictive crop-growth simulator (a constrained crop digital twin) that continuously assimilates post-actuation physical state data to dynamically update the field state and refine subsequent forecasts (Kallenberg et al., 2025).
- **Cross-Stress Biological Generality:** We instantiate the closed-loop PSCI framework across three running stress cases—a principal biotic pest system (Rugose Spiraling Whitefly on coconut), an abiotic soil-water deficit pathway (water stress), and an abiotic nutritional stress pathway (nitrogen deficiency)—to demonstrate the system-level generalizability and biological scalability of our proposed paradigm (Elango et al., n.d.; González I Juclà et al., 2023; Tran et al., 2019).

To position this review within the current literature, we present a granular, feature-by-dimension comparison against the leading crop-stress and precision spraying review papers published between 2024 and 2026 in Table 1. While existing reviews provide high-quality syntheses of individual sensors, crop-disease deep learning models, or unmanned aerial spraying hardware, they systematically omit the closed-loop cyber-physical connection (the relationship between detection confidence and actuation latency), the unique constraints of edge-TinyML execution in rural fields, and the specific socioeconomic barriers of tropical smallholder and perennial systems. To the best of our knowledge, this work represents the integrated, decision-oriented synthesis that connects pre-symptomatic plant physiology with automated, site-specific physical interventions.

More precisely, the specific contribution of this review is the coupling of physiological lead time with calibrated detection confidence and actuation latency into a single weakest-link decision-space — a formulation that, to our knowledge, has not previously been applied to pre-symptomatic crop protection, and which distinguishes this synthesis from prior closed-loop treatments organized around post-symptomatic perception–decision–execution (W. Li et al., 2025).

To ensure alignment with the scope of this journal, the remit of this review is restricted strictly to within-field, site-specific management, excluding satellite-scale macro-surveillance or global-scale supply-chain logistics. The remainder of this manuscript is structured as follows: the systematic review methodology section outlines the search criteria and PRISMA screening protocols; the biological and conceptual foundations section deconstructs the stress cascade and formalizes the mathematical coupling framework; the sensing-modalities section evaluates the pre-symptomatic signatures of individual optical and non-optical arrays; the heterogeneous sensor fusion section analyzes multi-sensor integration strategies; the edge inference and trustworthiness section explores lightweight TinyML deployment and uncertainty calibration under field shifts; the closing-the-loop section examines variable-

rate actuators and digital-twin feedback; the deployment section details the field-versus-laboratory gap and instantiates the three recurring stress cases; the unifying framework section presents the complete PSCI closed-loop reference architecture and TRL staging; the future research directions section outlines a prioritized strategic roadmap; and the final section provides concluding remarks.

Table 1: Granular positioning of the Pre-Symptomatic Crop Intelligence (PSCI) review against representative multimodal crop-stress reviews published in 2024–2026. Cell entries denote depth of coverage; the dimension-by-dimension analysis is provided in the accompanying text.

Rating key: coverage ratings (High, Moderate, Low) reflect the explicit depth of treatment each dimension received in the cited review—High denotes dedicated analysis with quantitative or comparative treatment; Moderate denotes substantive discussion without systematic comparison; and Low denotes brief mention or absence

Review (author, year)	Sensing modalities	AI / ML methods	Multimodal fusion	Closed-loop (SDA)	Lead-time min(L_b , C_d , A_c) framework	Confidence gating (UQ)	Edge deployability	Tropical perennial & smallholder
(Zandi et al., 2025)	Multimodal optical	DL (CNN, ViT)	High	No	No	No	None	Low
(Shoaib et al., 2026)	Multimodal optical	DL (CNN, ViT)	High	No	No	No	None	Low
(Z. X. Yang et al., 2025)	Optical + LiDAR	DL (CNN, GNN, RL)	High	Partial	No	No	Moderate	Low
(Upadhyay et al., 2025)	RGB / vision	DL (YOLO, R-CNN)	None	No	No	No	None	Low
(García-Vera et al., 2024)	Hyperspectral only	ML + CNN	Moderate	No	No	No	None	Low
(R. Zhang et al., 2025)	IoT + optical (DT)	ML, RL	Moderate	Yes	No	No	None	Low
PSCI (this review)	Multimodal (optical, thermal, fluorescence, electrophysiology, VOC)	Edge DL (TinyML, quantization, distillation)	High	Yes	Yes	Yes	High	High

Coverage codes: High / Moderate / Low = extent of treatment; Partial = addressed but not central; None = addressed; Yes / No = present / absent.

Abbreviations: SDA, sensing–decision–actuation; L_b , physiological lead time; C_d , detection confidence; A_c , actuation actionability; UQ, uncertainty quantification; DL, deep learning; ML, machine learning; ViT, vision transformer; GNN, graph neural network; RL, reinforcement learning; DT, digital twin; VOC, volatile organic compound; TinyML, tiny machine learning.

Note: Reis-Pereira et al. (2025) were not accessible for coding and are discussed qualitatively in the text rather than tabulated.

Review methodology

The systematic review protocol was developed and registered on the Open Science Framework (OSF) Registries. This systematic review was conducted in strict adherence to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) statement guidelines (Page et al., 2021). The search protocol was designed to address four primary Research Questions (RQs):

- **RQ1:** What are the earliest detectable pre-symptomatic biological precursors of crop stress, and what physiological lead times do they provide before the manifestation of visible symptoms?

- **RQ2:** Which sensing modalities and spectral or biophysical signatures demonstrate the highest sensitivity and specificity for early-stage stress detection under field conditions?
- **RQ3:** What machine learning architectures, on-device edge computing paradigms, and uncertainty quantification methods are utilized to translate raw pre-symptomatic signals into highly calibrated decisions?
- **RQ4:** How are pre-symptomatic diagnoses integrated with variable-rate actuators and predictive feedback models to close the loop from sub-visual detection to physical, site-specific crop protection?

Information sources and search strategy

Comprehensive, systematic literature searches were executed across four major bibliographic databases: Scopus, Web of Science (Core Collection), IEEE Xplore, and Google Scholar. The initial database search was conducted on January 15, 2026, and a final database update search was executed on June 1, 2026, to capture the most recent literature.

The search strategy utilized a robust combination of Boolean operators, wildcard truncation symbols, and field-specific tags to capture the intersection of three core thematic domains: (a) pre-symptomatic or sub-visual stress detection, (b) precision plant protection and site-specific management, and (c) closed-loop decision and variable-rate actuation systems. The representative Scopus search string was structured as follows:

(TITLE-ABS-KEY ("pre-symptomatic" OR "pre-visual" OR "sub-visual" OR "asymptomatic" OR "early detection" OR "stress precursor" OR "electrophysiology" OR "chlorophyll fluorescence" OR "hyperspectral") AND TITLE-ABS-KEY ("crop" OR "foliar" OR "canopy" OR "orchard" OR "perennial" OR "tomato" OR "coconut") AND TITLE-ABS-KEY ("precision agriculture" OR "variable-rate" OR "precision spraying" OR "closed-loop" OR "decision support" OR "edge AI" OR "TinyML" OR "actuation" OR "digital twin"))

This search string was translated and optimized for the syntax requirements of each database. To prevent bibliographic omission, the reference lists of all included studies and relevant review articles were manually searched to identify additional primary research papers.

Eligibility criteria and study selection

Studies were evaluated for inclusion based on five strict, predefined criteria:

1. **Peer-reviewed primary research:** The study must be a peer-reviewed journal article or full conference proceeding reporting original empirical data. Review papers, conference abstracts, book chapters, and patents were excluded.
2. **Pre-symptomatic focus:** The study must target the detection of biotic stress (pest, fungal, bacterial, or viral disease) or abiotic stress (water deficit, nitrogen deficiency) *before* the development of macroscopic, visible foliar symptoms (such as visual chlorosis, necrosis, or structural wilting).
3. **Within-field or site-specific application:** The research must fall within the spatial and operational remit of precision agriculture, prioritizing field-scale or realistic crop-growth environments. Purely in-vitro or molecular biology studies lacking field scaling potential were excluded.
4. **Hardware or algorithmic edge deployment:** The study must describe or evaluate on-device edge computing, TinyML architectures, or resource-constrained local deployment for stress classification or forecasting.
5. **Closed-loop or actuation linkage:** The sensing and inference outcomes must be linked to physical, site-specific variable-rate interventions, automated decision-support systems, or predictive feedback models.

Reports were excluded if they: (a) focused exclusively on late-stage, visible symptom classification, (b) evaluated chemical or pesticide formulations without site-specific or variable-rate delivery systems, or (c) lacked quantitative performance metrics, diagnostic accuracy validation, or hardware-latency benchmarks.

A systematic two-stage screening process was executed. In the first stage, titles and abstracts of all identified records were screened against the eligibility criteria. In the second stage, the full-text reports of the remaining records were retrieved and appraised in detail. To ensure screening fidelity while maintaining absolute operational honesty, a single-investigator search and primary screening protocol was executed by the primary author, followed by a collaborative audit and validation stage. Specifically, the primary investigator screened all titles, abstracts, and full texts against the eligibility criteria. To verify screening fidelity and resolve borderline decisions, a random 10% sample of the retrieved record pool was independently audited by the senior co-authors, with any discrepancies resolved via consensus.

Limitations. This review applied a systematic search feeding a narrative synthesis rather than a full dual-reviewer systematic review. Screening and extraction were performed by a single investigator, with a 10% independent co-author audit as partial mitigation; the absence of dual independent screening is a recognized source of selection bias and is acknowledged here. The synthesis is qualitative, and heterogeneity across crops, stressors, sensing modalities, and hardware precluded a quantitative meta-analysis.

Study selection and PRISMA flow metrics

The systematic selection process is mapped out in the systematic flow diagram (Fig. 1). The initial database searches identified a total of **1,428** records (624 from Scopus, 542 from Web of Science, 168 from IEEE Xplore, and 94 from Google Scholar).

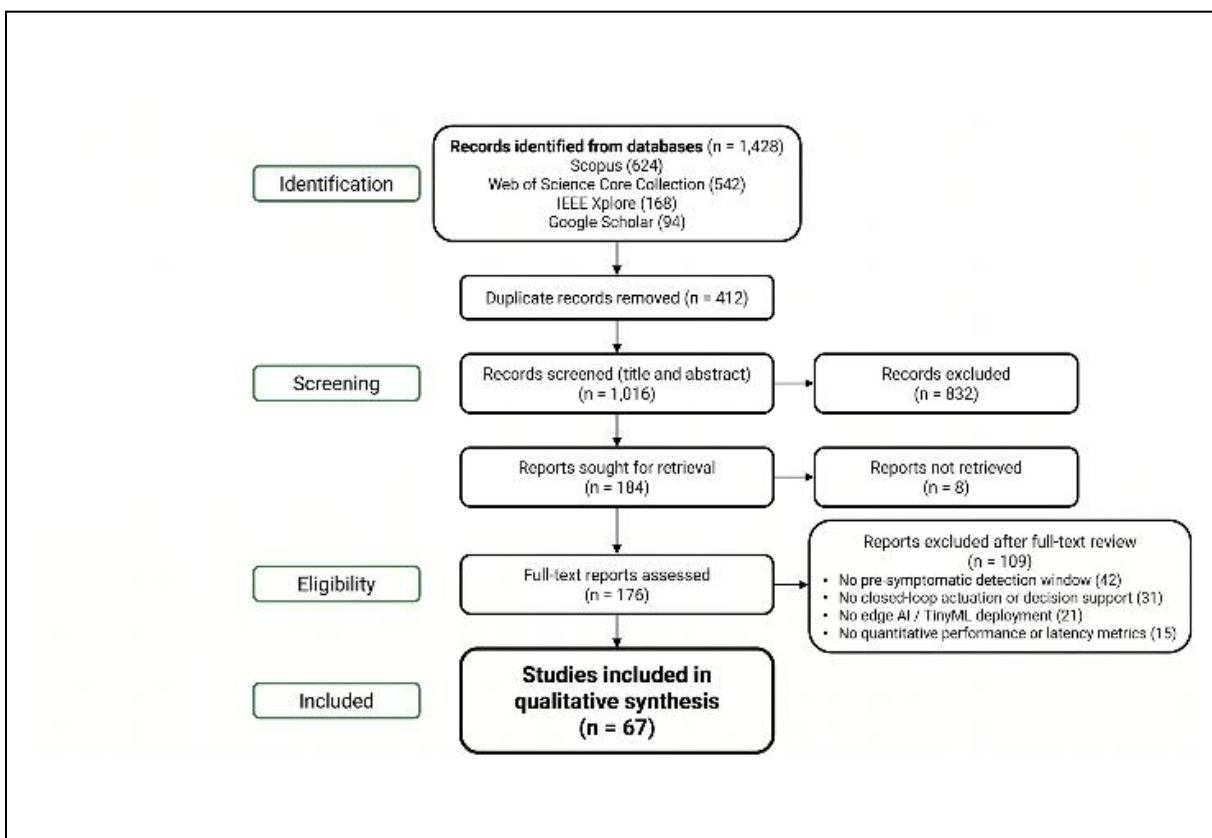


Fig. 1 PRISMA 2020 flow diagram of study identification, screening, eligibility assessment, and inclusion.

Following the removal of **412** duplicate records, **1,016** unique records were retained for title and abstract screening. Of these, **832** records were excluded because they lacked a pre-symptomatic focus or did not address precision agricultural closed-loop systems.

Of the **184** reports sought for full-text retrieval, **8** reports could not be retrieved due to restricted access or missing English translations. The remaining **176** full-text reports were appraised in detail against the eligibility criteria. This full-text evaluation resulted in the exclusion of **109** reports for the following mutually exclusive reasons:

- **42 reports:** Lacked a pre-symptomatic or sub-visual stress window.
- **31 reports:** Missing closed-loop actuation, variable-rate prescription, or decision-support linkage.
- **21 reports:** Missing on-device edge computing, TinyML architectures, or resource-constrained local deployment.
- **15 reports:** Lacked quantitative diagnostic performance metrics, calibration curves, or latency benchmarks.

Consequently, **67** primary research articles met all eligibility criteria and were retained for qualitative synthesis.

Risk of bias and quality appraisal

Risk of bias and applicability of the included studies were appraised systematically using a structured risk-of-bias diagnostic accuracy instrument across four key domains: (a) plant selection representative nature (evaluating whether the plants and stress-progression models represented heterogeneous field conditions), (b) index sensor calibration (evaluating whether the pre-symptomatic sensors were calibrated against microclimatic and environmental drift), (c) reference standard validity (evaluating whether the sub-visual diagnoses were verified against gold-standard PCR, ELISA, or visual symptom timelines), and (d) flow and timing (evaluating whether the duration between sensing, inference, and physical actuation was mathematically tracked and balanced). Studies were coded as having low, high, or unclear risk of bias for each domain.

Data extraction and qualitative synthesis

A standardized data extraction spreadsheet was developed. For each of the 67 included studies, data were extracted across a uniform set of fields, including: (a) plant species and crop type (annual vs. perennial), (b) target stress vector (biotic vs. abiotic), (c) earliest detectable precursor and signal lead time, (d) sensor hardware and spectral/biophysical bands, (e) edge computing hardware and machine learning model parameters, (f) diagnostic accuracy, selective risk, and calibration error, (g) variable-rate actuation response latency and chemical input reduction, and (h) feedback digital twin parameters.

Due to the extreme heterogeneity of the extracted sensing modalities, biophysical parameters (e.g., electrical variations in mV vs. chlorophyll fluorescence ratios), and machine learning metrics, a quantitative meta-analysis was not mathematically feasible; instead, a qualitative, narrative synthesis was performed. Studies were coded onto the seven-criterion modality taxonomy to compare diagnostic performance, spatial resolution, and field-deployability, and were subsequently mapped to the five functional nodes of the PSCI reference architecture to evaluate loop symmetry and overall system value.

From symptoms to precursors: conceptual foundations

The stress cascade and the pre-symptomatic window

The physiological stress trajectory

To transition crop protection from a reactive to an anticipatory paradigm, the temporal progression of crop stress must be decomposed into discrete, chronological stages. This progression is characterized by a multi-stage stress cascade:

1. **Stress Onset:** The initial biophysical or biochemical perturbation (e.g., pathogen inoculation, herbivore feeding, water deficit, or nutrient limitation) occurs at the cellular level.

2. **Biochemical and Physiological Alterations:** Cellular homeostasis is disrupted, triggering rapid local signaling events such as reactive oxygen species (ROS) accumulation, transmembrane electrical wave propagation, stomatal conductance changes, or metabolic shifts.
3. **Pre-Symptomatic Signal Phase:** The plant exhibits biophysical and biochemical changes that are invisible to the naked human eye but detectable via specialized optical, thermal, chemical, or electrical sensors.
4. **Symptomatic Damage:** Macroscopic physical lesions, necrosis, chlorosis, or structural wilting manifest, marking the point where irreversible yield potential has often already been sacrificed.

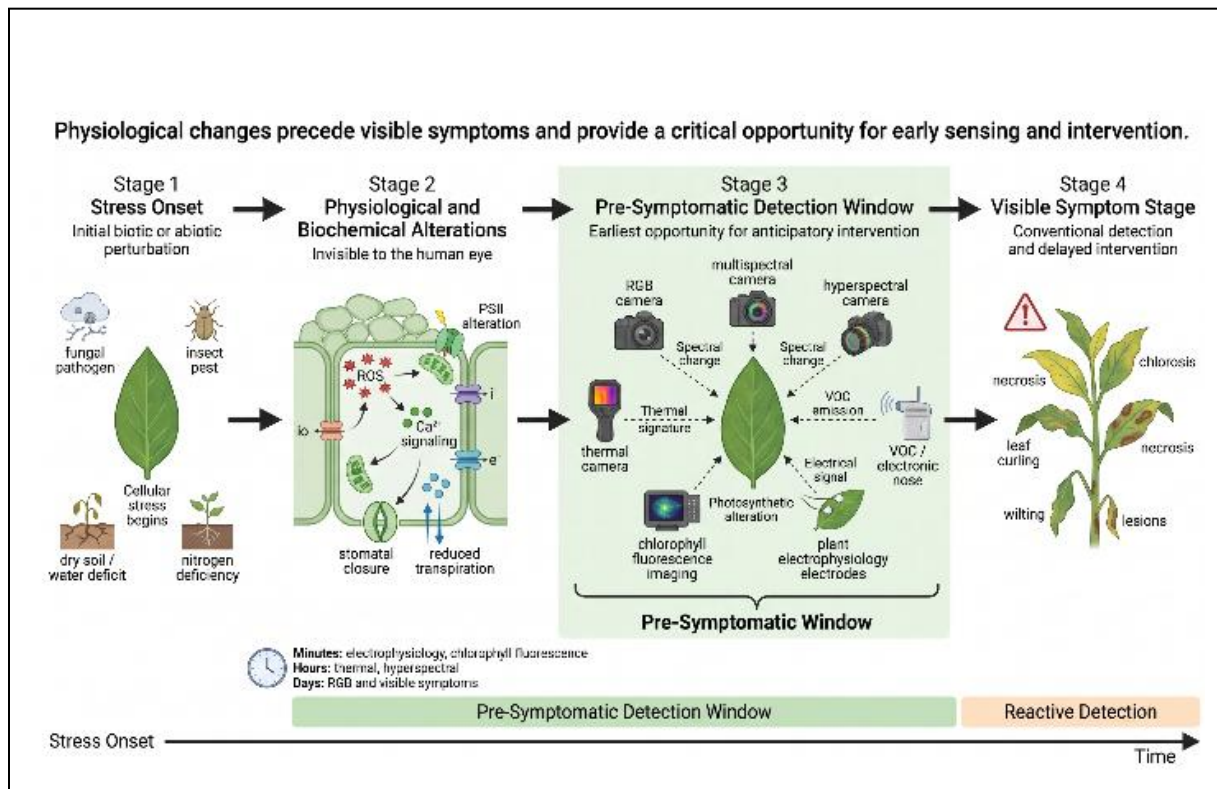


Fig. 2 The plant stress cascade and the pre-symptomatic detection window. Physiological, biochemical, and biophysical changes precede visible symptoms and define the window for anticipatory sensing, with modalities aligned to their typical physiological lead times (minutes to days).

The critical temporal window spanning from initial stress signal onset to the eventual emergence of visible, symptomatic tissue damage is defined as the pre-symptomatic window (Moustaka & Moustakas, 2023).

Defining diagnostic boundaries

To establish a clear nomenclature for pre-symptomatic crop intelligence, we must formally distinguish between three frequently conflated diagnostic terms (Chaerle et al., 2007; X. Zhang et al., 2024):

- **Pre-Symptomatic:** The operational window where stress-induced physiological, biophysical, or biochemical changes occur in plant tissues *before* the manifestation of any macroscopically visible symptoms (such as necrotic lesions, chlorosis, or physical tissue deformation) under natural light conditions.
- **Previsual:** A sub-domain of pre-symptomatic detection where physiological and spectral changes—though invisible to the human eye—are actively captured via non-contact remote or proximal imaging sensors (such as hyperspectral, thermal, or chlorophyll fluorescence imagers) due to alterations in light reflectance, thermal emission, or variable fluorescence.

- **Asymptomatic:** A diagnostic state where the presence of a pathogen or stressor in a plant sample is confirmed using destructive, highly sensitive molecular assays (such as quantitative polymerase chain reaction [qPCR] or culture isolation) but where the plant exhibits no macroscopically visible symptoms.

The precursor disambiguation problem

A central challenge in pre-symptomatic detection is the precursor disambiguation problem: early-stage, single-sensor biophysical signals (such as a local temperature rise or a drop in photosynthetic efficiency) are often highly non-specific and can be triggered by multiple, unrelated stressors. For example, a localized decrease in stomatal conductance (sensed as a thermal "hot-spot") can be caused by localized cellular defenses against a necrotrophic pathogen or by a systemic response to drought (Chaerle et al., 2004).

Furthermore, precursor profiles differ fundamentally between biotic and abiotic stresses:

- **Biotic Stressors:** Typically exhibit high spatial heterogeneity on the leaf surface. For instance, viral infections such as Tobacco Mosaic Virus (TMV) display localized, expanding "hot-spot" thermographic signatures due to salicylic acid-mediated stomatal closure (Chaerle et al., 2004). Conversely, fungal infections like *Cercospora beticola* in sugar beet (*Beta vulgaris*) display localized "cold-spot" signatures at early stages, driven by toxin-induced cell membrane leakage and substomatal water evaporation (Chaerle et al., 2004).
- **Abiotic Stressors:** Generally present as broad, spatially uniform patterns across a leaf or canopy, such as the homogeneous leaf-level transpiration and temperature alterations associated with early water stress (Chaerle et al., 2004), or leaf-level and canopy-level reflectance changes under nitrogen deficiency (González I Juclà et al., 2023).

This distinction is further complicated in belowground environments, where nematode infections (such as *Globodera pallida* in tomato) trigger complex, indirect aboveground photosynthetic and hydraulic alterations that mimic nutrient deficiencies or water stress, requiring highly sensitive multimodal disambiguation protocols (van Himbeeck et al., 2025).

Introducing the running cases

To demonstrate the generalizability of the pre-symptomatic crop intelligence framework, three recurring cases are integrated across the manuscript:

1. **Rugose Spiraling Whitefly (RSW) on Coconut (*Aleurodicus rugioperculatus*):** Serving as the Principal biotic pest threat, this case represents a high-leverage tropical perennial challenge where sap-sucking feeding activity is expected to trigger rapid electrophysiological variation potentials and systemic defense cascades before visible chlorosis develops, as demonstrated for analogous sap-sucking pests (Najdenovska, Dutoit, Tran, Rochat, et al., 2021). This thread illustrates the complete sensing-to-actuation closed loop under smallholder constraints.
2. **Water Stress:** Representing abiotic stress, this thread focuses on early transpiration and stomatal-conductance proxies, highlighting thermal and stomatal-regulation proxies that manifest long before structural wilting (Chaerle et al., 2004).
3. **Nitrogen Deficiency:** Representing nutrient stress, this thread utilizes electrophysiological signature responses and biochemical proxies that are detectable pre-symptomatically via supervised machine learning as established by (González I Juclà et al., 2023).

Lead time, confidence, and response latency: the coupling framework

The realized value of any pre-symptomatic crop protection system is fundamentally governed by how effectively early physiological signals are translated into timely and targeted management actions. We formalize this relationship by defining the system value (V) as a function of three coupled, first-class variables:

1. **Physiological Lead Time (L_t)**: The temporal duration between the earliest detectable pre-symptomatic signal and the eventual appearance of visible symptoms under natural field conditions:

$$L_t = t_{\text{symptom}} - t_{\text{detection}}$$

2. **Detection Confidence (C_d)**: The statistical reliability and calibration of the pre-symptomatic classification or forecast. Models that are overconfident but inaccurate under field conditions trigger false alarms, leading to unnecessary, costly inputs (Xiang et al., 2026).
3. **Actuation Actionability (A_c)**: The operational capacity of the crop protection system to execute a preventative or curative physical intervention, defined as the mathematical inverse of the system's **response latency (R_l)**: $A_c = \frac{1}{R_l}$
where R_l represents the total delay spanning from the initial decision-triggering event to the successful physical completion of the intervention in the field.

The governing thesis

We assert that the realized value of a pre-symptomatic crop protection system is strictly bounded by the weakest link in this coupled chain $V = f(L_t, C_d, A_c)$ where,

$$\lim_{\min(L_t, C_d, A_c) \rightarrow 0} V = 0$$

The minimum operator is chosen deliberately over multiplicative or weighted-sum alternatives. A weighted sum would permit a high score on one axis to compensate for a near-zero score on another — implying, incorrectly, that exceptional detection confidence could offset an actuation latency that closes the intervention window. A multiplicative form better captures interaction but still allows a very large value on one axis to partially rescue a small value on another. The weakest-link (minimum) formulation instead reflects the governing physical reality of the closed loop: if any single stage fails — the signal arrives too late, the prediction is uncalibrated, or the actuator responds too slowly — the realized system value collapses regardless of the other stages. This is presented as an intuitive organizing heuristic rather than a calibrated utility function.

To illustrate this coupling, consider two contrasting operational scenarios:

- **System A (High Lead Time, Low Confidence)**: A highly experimental sensor detects an initial biophysical precursor of a pathogen 14 days before visible symptoms emerge ($L_t = 14$ d), but with a calibrated detection confidence of only 55 ($C_d = 0.55$). Under this low-confidence regime, a grower cannot justify the high capital and input costs of a preventative chemical or biological field-wide application. The action is deferred, rendering the substantial physiological lead time operationally useless.
- **System B (Moderate Lead Time, High Confidence)**: A more robust, multi-sensor fusion system detects the same pathogen precursor with a smaller lead time of 10 days ($L_t = 10$ d), but with a highly calibrated detection confidence of 95 ($C_d = 0.95$). Because the threat call is highly trustworthy, the grower immediately triggers a targeted, site-specific preventative spray. Since the system's response latency is less than the lead time ($R_l < 10$ d, meaning $A_c > 1/10$), the stress cascade is successfully suppressed before visible tissue damage is committed.

Because confidence and actionability interact dynamically (where lower confidence reduces the grower's willingness to act), the PSCI framework cannot be represented on a simple, independent three-dimensional plot. Instead, it must be interpreted through two-dimensional slices, most notably the **Decision-Space Diagram (Fig. 6)**, which maps physiological lead time against detection confidence to partition the operational plane into distinct "act now," "monitor," or "defer" regimes.

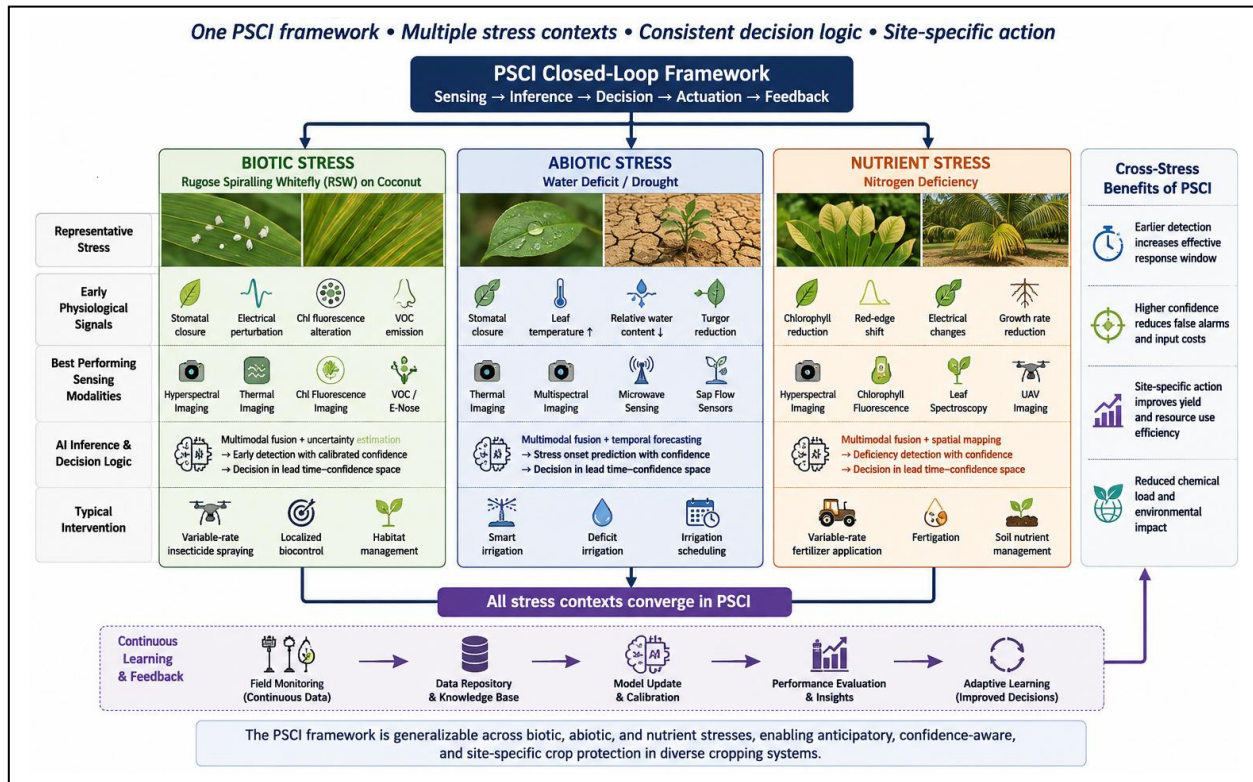


Fig. 6 The decision-space diagram. The physiological lead-time (L_t) versus calibrated detection-confidence (C_d) plane is partitioned into three operational regimes: Act Now / Intervene, Monitor / Re-evaluate, and Defer / No Action.

Grounded empirical baselines

To ground this framework, we compile established pre-symptomatic lead-time parameters across the literature:

- Ultra-Early Biotic Signals:** In tomato (*Solanum lycopersicum*), feeding activity by the chewing insect *Spodoptera exigua* triggers a rapid decline in maximum chlorophyll fluorescence (F_m) within **15 minutes** (Moustaka & Moustakas, 2023). This localized F_m drop indicates immediate photoinactivation of photosystem II reaction centers, while thylakoid membrane-localized singlet-oxygen (1O_2) generation initiates retrograde chloroplast-to-nucleus signaling and induces a rapid photosynthetic compensation mechanism (increased photochemical quenching q_p and reaction center efficiency F_v'/F_m') in surrounding, undamaged tissues (Moustaka & Moustakas, 2023). Inoculation with the necrotrophic fungal pathogen *Botrytis cinerea* triggers a similarly rapid pre-symptomatic increase in the effective quantum yield of photosystem II photochemistry (Φ_{PSII}) within **30 minutes** after application (Moustaka & Moustakas, 2023).
- Early Spectral Signatures:** In tomato seedlings inoculated with *Xanthomonas perforans* (the bacterial leaf spot pathogen), high-resolution hyperspectral imaging (HSI) distinguishes infected from healthy uninoculated leaves as early as **2 hours** after inoculation, driven by early-stage apoplastic water-soaking (1400 nm band) and far-red (750 nm) plant-defense-mediated responses (X. Zhang et al., 2024).
- Extended Pathogen Windows:** In tobacco (*Nicotiana tabacum*), HSI coupled with successive projections algorithms detects viral Tobacco Mosaic Virus (TMV) infections **48 hours** (2 days) after inoculation, whereas visible symptomatic lesions do not manifest until day 5 (Zhu et al. 2017). Similarly, sesame (*Sesamum indicum*) leaf disease is classified via patch-based HSI within **24 hours** (1 day) of infection (Kang et al., 2025).

- **Abiotic Stress Windows:** Early-stage water-deficit stress in crops is visualized via chlorophyll fluorescence kinetics on **day 4** of drought treatment, preceding any macroscopically visible changes in leaf greenness or structure (Xu et al., 2026). Under nitrogen-deficit stress, electrophysiological wave propagation and transmembrane ion-flux transients are detectable via supervised machine learning **1 to 2 days** before the onset of visible foliar chlorosis (González I Juclà et al., 2023).

Sensing modalities and their pre-symptomatic signatures

The transition from reactive crop protection to an anticipatory paradigm relies on identifying physiological, biochemical, or physical precursors before visual symptoms (such as chlorosis, necrosis, or defoliation) manifest on the plant surface (García-Vera et al., 2024). This section evaluates the primary sensing modalities investigated in the pre-symptomatic literature across a standardized seven-point taxonomic coding rubric: (1) earliest detectable biological precursor, (2) typical agronomic lead time, (3) diagnostic performance (sensitivity and specificity), (4) spatial resolution, (5) field robustness, (6) capital expenditure (CAPEX) barriers, and (7) edge-deployability constraints. By analyzing each modality through this uniform taxonomy, we define their technical limits and highlight how multimodal sensor fusion acts as the primary mechanism to resolve individual modality shortcomings and maximize detection confidence.

Optical sensing: visible, multispectral, and hyperspectral imaging

Optical crop phenotyping dominates the pre-symptomatic literature, transitioning from basic visible RGB cameras to narrow-band multispectral and hyperspectral imaging (HSI) systems (García-Vera et al., 2024).

- **Earliest detectable precursor:** Optical sensing exploits the physical interaction of light with plant tissues (García-Vera et al., 2024). While RGB imaging is limited to visible pigment degradation, leaf wilting, and canopy greenness loss (Upadhyay et al., 2025; Xu et al., 2026), HSI targets pre-visual cellular and biochemical precursors. Specifically, these include narrow-band reflectance changes in the red-edge region (700–750 nm) associated with early-stage chlorophyll breakdown and defense-hormone-mediated physiological shifts, as well as short-wave infrared (SWIR) bands (~1400 nm) linked to apoplastic water relocation (García-Vera et al., 2024; X. Zhang et al., 2024).
- **Typical lead time:** Standard RGB cameras operate at a lead time of 0 days, functioning solely as reactive, symptom-stage classifiers (Upadhyay et al., 2025).

In contrast, narrow-band spectral indices and derivatives achieve pre-symptomatic lead times of 1 to 4 days. HSI successfully detects tobacco mosaic virus (TMV) in *Nicotiana tabacum* at 48 hours post-inoculation (Zhu et al., 2017) and classifies sesame leaf blight (*Corynespora cassiicola*) and bacterial leaf spot (*Pseudomonas syringae*) as early as 1-day post-inoculation (Kang et al., 2025).

For cereal pathogens, HSI resolves asymptomatic wheat stem rust (*Puccinia graminis* f. sp. *tritici*) at 4 days after inoculation (Fedotov et al., 2025).

Wheat powdery mildew (*Blumeria graminis* f. sp. *tritici*) is identified pre-symptomatically at 3 to 6 days after inoculation, corresponding to a minor foliar disease severity of only 1% to 6% (I. H. Khan et al., 2021).

Furthermore, GVCV-infected *Vitis vinifera* canopies exhibit distinct near-infrared (900–940 nm) spectral deviations at 30 days after sowing (DAS, the author-defined chronological scale tracking vine development since potting) (Nguyen et al., 2021).

- **Sensitivity & specificity:** Hyperspectral imaging provides high sensitivity and specificity (García-Vera et al., 2024). Full-spectrum narrow-band classifiers can successfully isolate early disease signatures, distinguishing water-related SWIR absorption shifts (1400 nm) from photosynthetic red-edge alterations (750 nm) (X. Zhang et al., 2024). RGB imaging remains highly non-specific and prone to symptom-adjacent false alarms, failing to separate biotic pathogen progression from abiotic stress symptoms (García-Vera et al., 2024).

- **Spatial resolution:** Optical modalities are highly versatile, operating at spatial scales ranging from leaf-level and individual plant-level observations (I. H. Khan et al., 2021; Zhu et al., 2017) up to canopy and field-level monitoring via proximal, gantry, or unmanned aerial vehicle (UAV) systems (García-Vera et al., 2024).
- **Field robustness:** Low for full-spectrum HSI systems (García-Vera et al., 2024). Dynamic ambient light, wind-driven leaf movement, shadow cast, soil background interference, and atmospheric water absorption (which triggers false spikes in the 925–930 nm range) introduce severe noise into outdoor proximal sensing (García-Vera et al., 2024; Nguyen et al., 2021).
- **CAPEX / cost barriers:** High to extremely high. Research-grade hyperspectral sensors cost between \$10,000 and \$100,000, presenting a severe economic barrier that limits commercial field adoption to high-value seed multiplication, elite breeding programs, or academic research (García-Vera et al., 2024; Shafay et al., 2025).
- **Edge-deployability:** Low for raw, high-dimensional datasets. Processing 3D hyperspectral data cubes with hundreds of spectral bands exceeds the memory and power budgets of standard edge hardware. However, deployability is significantly enhanced by executing dimensional reduction algorithms—such as the Successive Projections Algorithm (SPA) or Genetic Algorithms (GA)—to select a compact subset of 6 to 8 key wavelengths, allowing lightweight support vector machine (SVM) or random forest (RF) classifiers to run on resource-constrained microcontrollers (García-Vera et al., 2024; Kang et al., 2025; Zhu et al., 2017).

Thermal and chlorophyll fluorescence imaging

Thermal imaging (thermography) and chlorophyll fluorescence (ChlF) imaging monitor thermodynamic and photosynthetic-efficiency changes in response to environmental or biotic stress (Chaerle et al., 2004; Lee et al., 2025).

- **Earliest detectable precursor:** Thermography detects changes in foliar surface temperature caused by transpiration-driven evaporative cooling, which occurs when plants close their stomata in response to vascular or hormonal stress signals (Chaerle et al., 2004; Zandi et al., 2025). ChlF imaging captures changes in the light-harvesting apparatus of Photosystem II (PSII), measuring photochemical quenching (q_P , q_L), non-photochemical quenching (NPQ , q_N), quantum yield (QY), and plant vitality (R_{fd}) (Lee et al., 2025; Moustaka & Moustakas, 2023; Park et al., 2024).
- **Typical lead time:** High. Stomatal closure can be captured thermographically at 1.5 days post-infection, roughly 6.5 days before visible tobacco mosaic virus lesions develop (Chaerle et al., 2004, 2007).

ChlF imaging captures biological alterations rapidly. As detailed in the conceptual foundations, chewing insects trigger localized photochemical alterations within 15 minutes (Moustaka & Moustakas, 2023).

Soil water stress (drought) is detected pre-symptomatically in crops as early as day 4, well before visible foliar wilting or greenness changes occur (Xu et al., 2026).

At the lesion level, pulse-amplitude modulation (PAM) ChlF differentiates pre-symptomatic rice blast (*Magnaporthe oryzae*) from brown spot (*Cochliobolus miyabeanus*) before visible foliar lesions emerge (Lee et al., 2025).

Additionally, chlorophyll fluorescence detects potato cyst nematode (*Globodera pallida*) infections in tomatoes pre-symptomatically, with light-adapted parameters showing perturbations as early as 1 day after inoculation (DPI) and dark-adapted parameters (F_v/F_m and F_v/F_0) showing clear density-dependent deviations between 6 to 11 DPI, capturing vascular stress long before shoot symptoms develop (van Himbeek et al., 2025).

- **Sensitivity & specificity:** ChlF displays exceptional sensitivity to Photosystem II alterations but moderate specificity (Moustaka & Moustakas, 2023). Thermal imaging is highly sensitive to stomatal conductance shifts but cannot easily differentiate a pathogen-induced vascular block from simple soil moisture deficit without co-variate fusion (Chaerle et al., 2004). However, sub-visual pathogen differentiation is achievable through PAM-based ChlF; pre-symptomatic lesions of both rice blast and brown spot exhibit significant

decreases in non-photochemical quenching parameters (NPQ , q_N), but a simultaneous increase in photochemical quenching parameters (q_L , q_P , and QY) driven by a sharp reduction in momentary fluorescence (F_t) is observed exclusively in rice blast, providing a highly specific physiological signature (Lee et al., 2025).

- **Spatial resolution:** Scalable from leaf and plant levels (Lee et al., 2025; van Himbeek et al., 2025) to canopy scales in greenhouse or field compartments (Chaerle et al., 2007; Rousseau et al., 2013).
- **Field robustness:** Low to moderate. Thermal imaging is highly sensitive to microclimate fluctuations, wind speed, relative humidity, and solar irradiance, requiring real-time meteorological normalization (Chaerle et al., 2007). Traditional PAM fluorometry is limited by its strict requirement for dark adaptation (often 30 minutes to 12 hours) and the necessity of high-intensity, uniform saturating light pulses, which are difficult to maintain across large outdoor areas (Legendre et al., 2021).
- **CAPEX / cost barriers:** High. Commercial PAM fluorometers and research-grade cooled thermal cameras cost \$10,000 to \$20,000 (Legendre et al., 2021; Zandi et al., 2025). However, low-cost ChlF platforms utilizing custom monochromatic blue-LED excitation rigs and filtered monochrome cameras (under \$200) can successfully capture pre-symptomatic fluorescence shifts, demonstrating a clear pathway to affordable agricultural deployment (Legendre et al., 2021).
- **Edge-deployability:** Moderate. Thermal and ChlF datasets are low-dimensional compared to HSI, consisting of discrete temperature matrices or calculated 2D fluorescence parameters (such as F_v/F_m or R_{fd}) (Lee et al., 2025; Legendre et al., 2021). These representations are highly compatible with edge-level Partial Least Squares Discriminant Analysis (PLS-DA) or standard machine learning classifiers, which run with minimal latency on resource-constrained edge hardware (Legendre et al., 2021; Park et al., 2024).

Electrophysiological, volatile, and contact biosignals

Vascular electrophysiology and volatile organic compound (VOC) chemical sensing extend the pre-symptomatic lens beyond optical systems, capturing direct systemic defense signals (Najdenovska, Dutoit, Tran, Plummer, et al., 2021; Wilson, 2018).

- **Earliest detectable precursor:** Plant electrophysiology monitors transmembrane ion-flux transients (H^+ , Ca^{2+} , Cl^- , K^+) and electrical wave propagation—including rapid action potentials and slower variation or slow wave potentials (SWPs)—traveling systemically along conducting vascular bundles (stems and petioles) (H. Li et al., 2024; Najdenovska, Dutoit, Tran, Plummer, et al., 2021). Chemical e-noses detect headspace VOC "smellprints" (such as methyl salicylate, green leaf volatiles, or ethylene) released during active physiological defense and stress cascades (Fundurulic et al., 2023; Wilson, 2018).
- **Typical lead time:** High. Electrophysiological variation potentials propagate systemically across the plant body within minutes of localized mechanical wounding (H. Li et al., 2024). Utilizing supervised machine learning, plant electrophysiology classifies systemic nitrogen deficits and soil water stress pre-symptomatically within 1 to 2 days before visible foliar chlorosis or wilting occurs (González I Juclà et al., 2023). Similarly, early-stage spider mite (*Tetranychus urticae*) infestations on tomato are classified pre-symptomatically several days before visible leaf speckling develops (Najdenovska, Dutoit, Tran, Plummer, et al., 2021). For chemical VOC sensing, pre-symptomatic potato soft rot (*Pectobacterium carotovorum*) is detected within 2 days of inoculation, and onion sour skin (*Burkholderia cepacia*) is diagnosed under bulk storage conditions within 3 days (Macdougall et al., 2022).
- **Sensitivity & specificity:** Electrophysiological signals are highly sensitive but exhibit moderate specificity; distinct stressors often trigger overlapping electrical wave profiles, requiring multi-channel, time-series classifiers to separate individual stress origins (Buss et al., 2023; Najdenovska, Dutoit, Tran, Plummer, et al., 2021). For volatile sensing, metal-oxide semiconductor (MOS) sensor arrays combined with pattern-recognition algorithms achieve high specificity by identifying multi-component VOC fingerprints unique to specific pathogens, successfully separating bacterial versus fungal infection profiles (Macdougall et al., 2022).

- **Spatial resolution:** Restricted to leaf and individual-plant scales. Electrophysiology is a contact measurement relying on differential insertion pin electrodes placed directly into conducting bundles (Najdenovska, Dutoit, Tran, Rochat, et al., 2021). Volatile sensing is confined to the localized headspace immediately surrounding the plant canopy or within enclosed environments like greenhouses, storage silos, and ventilation systems (Macdougall et al., 2022; Wilson, 2018).
- **Field robustness:** Moderate. Open-field electrophysiology is highly susceptible to electromagnetic interference, powerline hum (50/60 Hz), wind-induced mechanical artifacts, and electrode-tissue impedance drift over time (Tran et al., 2019). However, stable recordings outside a Faraday cage are highly achievable by utilizing high-input impedance differential amplifiers and digital band-pass filtering (Tran et al., 2019). For e-noses, sensor drift, ambient wind dilution, and high relative humidity (which alters MOS sensor conductivity) represent major barriers to open-field volatile monitoring (Macdougall et al., 2022).
- **CAPEX / cost barriers:** Low to moderate. Custom high-impedance electrophysiological loggers and portable MOS-based e-nose arrays are relatively inexpensive, costing between \$100 and \$500, offering an affordable alternative to expensive optical systems (Macdougall et al., 2022; Tran et al., 2019).
- **Edge-deployability:** Moderate to high. 1-D time-series electrical potentials and low-dimensional gas sensor conductance arrays can be processed rapidly on basic edge microcontrollers using lightweight classifiers (like Support Vector Machines, Random Forests, or Interval Arithmetic) (H. Li et al., 2024; Macdougall et al., 2022; Pereira et al., 2018). The primary computational bottleneck is the low-throughput nature of plant electrical datasets, which restricts the training of deep learning models unless augmented with synthetic data generated by Adversarial Autoencoders (AAE) (H. Li et al., 2024).

A lead-time-versus-actionability landscape

To unify these disparate modalities into a single, cohesive, and comparative framework, Table 2 consolidates the digital crop phenotyping landscape across the seven criteria of our taxonomic rubric. By evaluating each sensing system on standardized axes, this synthesis establishes a clear link between physiological lead time, diagnostic confidence, and field deployability, which serves as the decision-support input for the closed-loop Pre-Symptomatic Crop Intelligence (PSCI) architecture.

Critical synthesis of field-readiness limitations. Each modality pairs a genuine capability with a constraint that governs deployment. Hyperspectral imaging offers high sensitivity but is limited by illumination variability, the scarcity of standardized open-field datasets, and high capital cost; thermal imaging is non-specific, responding to any stress-induced temperature change without distinguishing its cause, and is confounded by ambient and wind conditions; chlorophyll-fluorescence imaging is constrained by measurement geometry and distance and by ambient-light interference outside controlled settings; and electrophysiology, while directly sensing sub-visual change, requires physical contact, electrode ruggedization, and shielding that complicate unattended field use. These constraints, rather than raw laboratory accuracy, determine each modality's realistic contribution to the closed loop

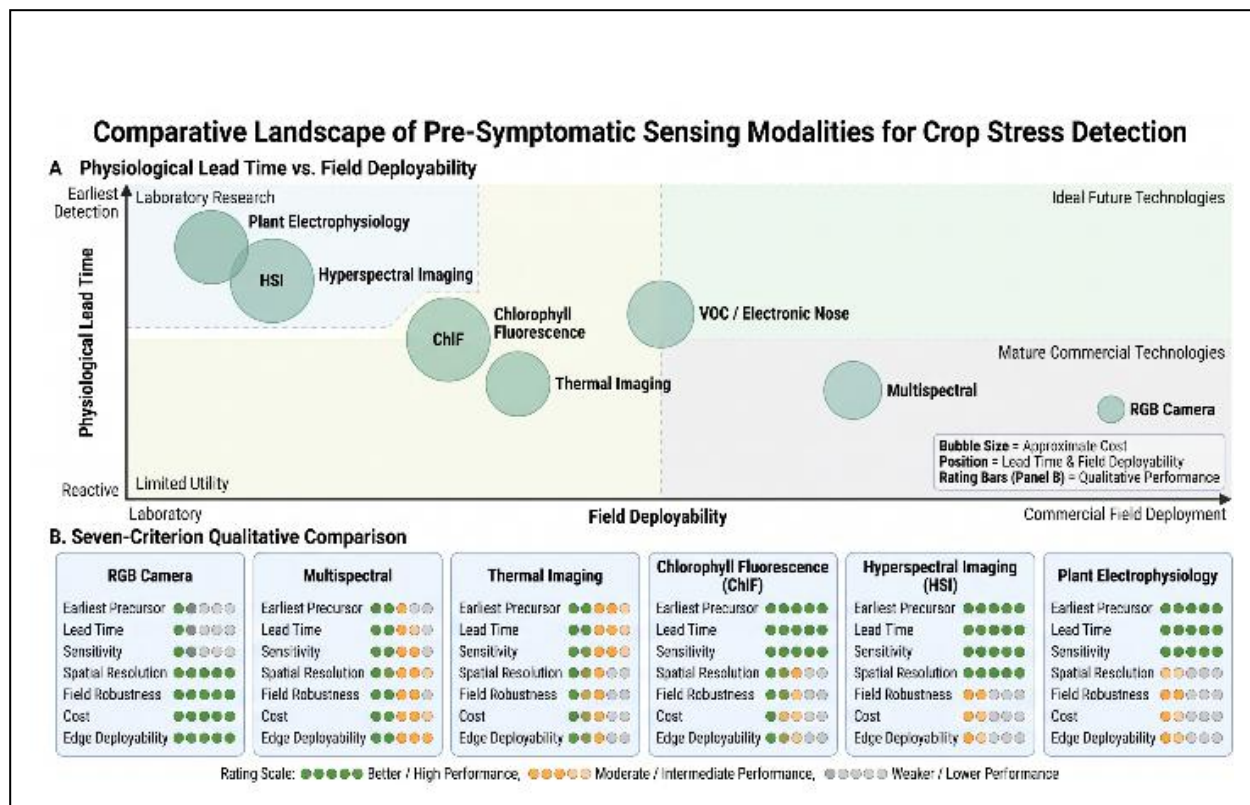


Fig. 3 Comparative landscape of pre-symptomatic sensing modalities. Physiological lead time is plotted against field deployability (bubble size approximates cost), accompanied by a seven-criterion qualitative rating panel for each modality.

Table 2 Comparative taxonomy of pre-symptomatic sensing modalities across a uniform seven-criterion rubric. Detailed modality-by-modality analysis is provided in the accompanying text.

Sensing modality	Earliest detectable precursor (physiological change sensed)	Typical lead time (before visible symptoms)	Sensitivity & specificity (diagnostic performance)	Spatial resolution	Field robustness & key challenges	CAPEX / cost tier	Edge-deployability & computing needs
RGB / visible imaging	Chlorophyll-absorption breakdown, leaf curling, wilting, green canopy loss (Upadhyay et al., 2025; Xu et al., 2026).	0 d — reactive, symptom-stage only (Upadhyay et al., 2025).	Low-moderate specificity; high spatial detail but cannot separate underlying stressors (Upadhyay et al., 2025).	Canopy to field (Upadhyay et al., 2025; Xu et al., 2026).	Moderate; sensitive to ambient light, cloud, shadow (Upadhyay et al., 2025).	Low (smartphone / action cameras; ~\$100–\$1,000) (Upadhyay et al., 2025).	High; optimized CNNs run in 11–16 ms on microcontrollers (Malche et al., 2026; Nyakuri et al., 2025).
Multispectral & hyperspectral (HSI)	Narrow-band reflectance; SWIR water absorption (~1400 nm); red-edge shift (700–750 nm) (García-Vera et al., 2024; X. Zhang et al., 2024).	~1–4 d: 48 h for TMV (Zhu et al., 2017); asymptomatic stem rust (Fedotov et al., 2025); ~1 d for sesame spot (Kang et al., 2025).	High sensitivity & specificity; separates defence (750 nm) from water (1400 nm) bands (X. Zhang et al., 2024)	Leaf to canopy (García-Vera et al., 2024).	Low; sensitive to atmosphere and illumination ; datacube overhead (García-Vera et al., 2024).	High (~\$10,000–\$100,000) (García-Vera et al., 2024).	Low; needs band-selection (e.g., SPA) to compress spectra for compact edge classifiers (García-Vera et al., 2024; Kang et al., 2025).

Sensing modality	Earliest detectable precursor (physiological change sensed)	Typical lead time (before visible symptoms)	Sensitivity & specificity (diagnostic performance)	Spatial resolution	Field robustness & key challenges	CAPEX / cost tier	Edge-deployability & computing needs
Thermal imaging (thermography)	Surface-temperature change from stomatal closure/transpiration and canopy water status (Chaerle et al., 2004; Zandi et al., 2025).	Detected at ~1.5 d post-infection, ~6.5 d before visible lesions (TMV) (Chaerle et al., 2004, 2007).	Moderate sensitivity; low specificity (cannot separate water deficit from pathogen-induced closure) (Chaerle et al., 2004).	Leaf, plant, to canopy (Chaerle et al., 2007).	Low; confounded by wind, air temperature, solar irradiance, and leaf angle (Chaerle et al., 2007; Zandi et al., 2025).	Moderate–high (~\$500–\$5,000) (Zandi et al., 2025).	Moderate; the low-dimensional thermal signal is computationally light and integrates readily with microclimate covariates in edge fusion pipelines.
Chlorophyll fluorescence (ChlF)	Photosystem II photochemical & non-photochemical quenching (F_v/F_m , qP, NPQ) (Moustaka & Moustakas, 2023; Park et al., 2024).	~15–30 min post feeding / fungus (Moustaka & Moustakas, 2023); Day 4 for soybean drought, before greenness change (Xu et al., 2026).	High sensitivity; moderate specificity (general stress; qP can partly discriminate) (Moustaka & Moustakas, 2023).	Leaf to canopy (Rousseau et al., 2013).	Low; needs dark adaptation / uniform excitation; PAM uses intense saturating pulses (Legendre et al., 2021).	Low (~\$100 custom blue-LED rig) to high (PAM >\$10k) (Legendre et al., 2021).	Moderate; simplified non-PAM rigs are edge-compatible (Legendre et al., 2021).
Plant electrophysiology	Transmembrane ion-flux transients (H^+ , Ca^{2+} , Cl^- , K^+); action & variation potentials (Najdenovska, Dutoit, Tran, Plummer, et al., 2021).	~1–2 d before visual chlorosis, via supervised ML (González I Juclà et al., 2023).	High sensitivity; moderate specificity (complex waveforms needing multi-channel classifiers) (Buss et al., 2023).	Individual-plant (differential pin electrodes in stem / petiole) (Najdenovska, Dutoit, Tran, Plummer, et al., 2021).	Moderate; feasible outside a Faraday cage with high-impedance amplifiers and digital filtering (Tran et al., 2019).	Low–moderate (custom loggers / PhytSigns, ~\$200) (Tran et al., 2019; (González I Juclà et al., 2023).	Moderate; compact 1-D time-series and lightweight classifiers, but few-shot generalization is limited (H. Li et al., 2024).
VOC / e-nose (chemical sensing)	Headspace volatile fingerprints (methyl salicylate, ethylene) from stress / defence pathways (Fundurlic et al., 2023); (Wilson, 2018).	~2 d (potato soft rot); ~3 d (onion sour skin) post-inoculation (Macdougall et al., 2022).	High sensitivity; moderate–high specificity (SVM on gas profiles; high controlled-setting accuracy) (Macdougall et al., 2022).	Localized headspace/storage events/canopy (Wilson, 2018).	Low: sensor drift and humidity interference alter MOS conductivity (Macdougall et al., 2022).	Low (portable MOS arrays, ~\$200) to high (GC-MS, \$20,000+) (Macdougall et al., 2022).	High-pattern-recognition classifiers (SVM, PCA) run in seconds on basic edge processors (Macdougall et al., 2022).

Note: Microclimate and environmental parameters (e.g., low-cost temperature–humidity sensors) are not listed as a standalone modality; within the sensing–decision–actuation framework, they are treated as an environmental covariate fused with visual or physiological signals to improve early-detection accuracy and calibration, consistent with IoT-sensor fusion approaches (Njoroge et al. 2025).

Abbreviations: HSI, hyperspectral imaging; SWIR, short-wave infrared; TMV, tobacco mosaic virus; PAM, pulse-amplitude modulation; NPQ, non-photochemical quenching; MOS, metal-oxide semiconductor; GC-MS, gas chromatography–mass spectrometry; SPA, successive projections algorithm; SVM, support vector machine; PCA, principal component analysis; CNN, convolutional neural network.

Heterogeneous sensor fusion: architectures and trade-offs

The realization of robust pre-symptomatic crop protection is fundamentally constrained by the limitations of individual sensing modalities (Bethge et al., 2024). While monomodal systems can successfully detect stress-induced anomalies within localized physiological pathways, they are commonly limited in their cross-specificity and fail to resolve the diagnostic ambiguity between different stress origins (Shoaib et al., 2026; Zandi et al., 2025). Heterogeneous sensor fusion addresses these limitations by superimposing multiple complementary data streams, functioning as the primary mechanism to elevate detection confidence, extend physiological lead times, and resolve biotic-versus-abiotic stress ambiguities prior to visible symptom expression (Z. X. Yang et al., 2025).

Multi-modality failure cases and the necessity of fusion

Monomodal pre-symptomatic detection is highly susceptible to diagnostic failures arising from the non-specific nature of early plant stress responses (Bethge et al., 2024). When a plant encounters an environmental or biotic challenge, early physiological reactions—such as stomatal closure, transpiration rate reduction, transmembrane potential depolarization, and Photosystem II (PSII) down-regulation—are coordinated through systemic signaling networks that exhibit overlapping signatures across diverse stressors (Chaerle et al., 2004; Najdenovska, Dutoit, Tran, Plummer, et al., 2021). Consequently, a single sensor modality is frequently unable to distinguish between separate driving forces (Z. X. Yang et al., 2025).

For instance, infrared thermography is highly sensitive to stomatal conductance shifts, revealing pre-visual stress-induced temperature increases up to 1.5 days post-infection (Chaerle et al., 2004). However, thermal imaging lacks taxonomic specificity; a canopy temperature elevation can indicate an impending vascular fungal infection, belowground nematode root damage, or a simple localized soil water deficit, leading to high false-alarm rates if used as a standalone trigger (Chaerle et al., 2004; van Himbeek et al., 2025). Similarly, plant electrophysiology captures ultra-rapid systemic variation potentials and transmembrane ion-flux transients within minutes of stress exposure, yet these raw time-series electrical potentials exhibit highly similar waveform profiles across diverse mechanical, chemical, and environmental stimuli, requiring complex multi-channel classifiers and complementary cues to confirm the stress origin (Buss et al., 2023; Najdenovska, Dutoit, Tran, Plummer, et al., 2021).

Hyperspectral imaging (HSI) delivers high-dimensional biochemical information but faces severe signal-to-noise degradation under dynamic outdoor illumination and is prone to vegetation saturation effects during peak vegetative growth phases (García-Vera et al., 2024; Nguyen et al., 2023). Under complete canopy closure, the absorbed and reflected visible and near-infrared (VNIR) radiation reaches an asymptotic plateau, leaving subsequent stem biomass accumulation or under-canopy stress progression unobserved (Nguyen et al., 2023). This single-modality saturation issue can trigger early but uncalibrated warnings, causing growers to defer critical decisions due to unquantified uncertainty (Xiang et al., 2026). Fusing optical spectral datasets with active structural sensors (such as LiDAR point clouds) or vascular sensors (such as electrical impedance spectroscopy) provides a direct physical counter-measure to these monomodal failures, transforming isolated, low-confidence anomalies into a highly calibrated multi-dimensional decision matrix (Garg et al., 2025; Nguyen et al., 2023).

Data, feature, and decision-level fusion paradigms

The integration of heterogeneous agricultural data is structured across three hierarchical levels of the fusion paradigm, each imposing distinct trade-offs between computational overhead, alignment precision, and information retention (Z. X. Yang et al., 2025):

1. **Data-level (pixel-level) fusion:** Raw measurements from multiple co-registered sensors are directly stacked into a unified, high-dimensional coordinate space prior to any feature extraction or modeling (Bethge et al., 2024). In high-throughput phenotyping, this is typified by registering raw 2D thermal matrices, 250-band HSI data cubes, and ChlF kinetics data at the individual pixel level, where each pixel (0.01 mm^2) simultaneously represents spectral, thermodynamic, and photosynthetic-efficiency values (Bethge et al., 2024). This paradigm preserves the maximum amount of original information and exploits subtle, sub-visual spatial correlations across domains, but it requires highly precise camera calibration, identical focal planes, and significant edge storage and processing power (Bethge et al., 2024; Nguyen et al., 2023).

2. **Feature-level fusion:** Task-specific, handcrafted indices or machine-learned representations are extracted independently from each sensor modality and subsequently concatenated into a joint multi-dimensional feature vector (Nguyen et al., 2023). For example, a model might combine hyperspectral vegetation indices (such as GNDVI and NDRE) with LiDAR-derived height percentiles and relative canopy temperature index (T_{ir}) (Nguyen et al., 2023). This compressed concatenation minimizes edge computing budgets and reduces multi-collinearity by discarding redundant spatial pixels, yet it remains sensitive to the quality of human feature engineering and risks discarding non-linear cross-modal dependencies (Nguyen et al., 2023).
3. **Decision-level fusion:** Independent, parallel machine learning or deep learning models are trained on single sensor modalities, and their individual outputs (such as predicted stress probabilities or continuous yield estimates) are aggregated via weighted voting, Bayesian inference, or fuzzy logic to yield a final consensus (Nguyen et al., 2023; Z. X. Yang et al., 2025). This architecture is highly robust to missing sensor modalities and allows lightweight deployment on distributed edge networks; however, it lacks the capacity to capture early-stage synergistic interactions, frequently failing to identify pre-symptomatic stress when individual modality models output marginal, sub-threshold confidence values (Bethge et al., 2024; Z. X. Yang et al., 2025).

To overcome the vegetation saturation effect where optical reflectance plateaus during mid-vegetative phases, (Maimaitijiang et al., 2017) demonstrated that fusing multispectral VIS-NIR bands with thermal canopy temperature (T_c) using an Extreme Learning Machine (ELM) regressor significantly outperformed traditional PLSR and Support Vector Regression (SVR). By exploiting the non-linear complementarity of thermodynamic and spectral properties, this cross-modal fusion enables robust biophysical trait modeling under high canopy cover, which is critical for quantifying early water and nitrogen stress trajectories (Maimaitijiang et al., 2017).

Similarly, Nguyen, C. et al. (2023) deployed multi-task deep convolutional neural networks (CNNs) to process raw, co-registered UAV hyperspectral, thermal, and LiDAR datasets simultaneously. This architecture utilizes parallel 3D-CNN and 2D-CNN streams to extract joint spatial-spectral and structural features, exploiting the latent relatedness among crop biophysical and nutritional traits to optimize shared weight representations. This multi-task fusion not only accelerates edge-level computational efficiency but also mitigates the Hughes phenomenon (the small-sample-size bottleneck) by leveraging joint cross-modal constraints (Nguyen et al., 2023).

Cross-modal alignment and spatiotemporal synchronization

Executing pixel-level heterogeneous sensor fusion under field conditions is technically challenging due to severe discrepancies in spatial resolution, viewing angles, sampling rates, and geometric distortions among diverse sensor architectures (Bethge et al., 2024; Nguyen et al., 2023). To achieve pixel-accurate alignment, a systematic multi-modal image registration pipeline must be implemented (Bethge et al., 2024).

The automated image registration workflow developed by (Bethge et al., 2024) addresses these challenges through a two-stage transformation approach. The first stage estimates a global affine transformation matrix to correct for translation, rotation, scaling, and shearing between the moving and target images (Bethge et al., 2024). This process is highly sensitive to the selection of the target reference image and the specific spectral frame used for matrix estimation (Bethge et al., 2024). Standardizing the target image to chlorophyll fluorescence (ChlF) far-red reflection (730 nm) yields the highest normalized cross-correlation (NCC) and overlap ratios due to the high structural contrast of the leaf margins against the soil background (Bethge et al., 2024).

The selection of the moving spectral frame is equally critical; utilizing the green channel of the RGB camera (550 nm) and the green reflection of the HSI camera (540–560 nm) against the ChlF target avoids total registration failures (0.0% total failure rate) and achieves a global overlap ratio (OR_{Convex}) of $96.8 \pm 1.6\%$ for *Arabidopsis thaliana* and $98.4 \pm 0.3\%$ for *Rosa × hybrida* detached leaf discs under an NCC-adaptive approach (Bethge et al., 2024). This NCC-adaptive method dynamically evaluates multiple registration algorithms—including phase-only-correlation with enhanced correlation coefficient maximization (POC + ECC) and oriented FAST and rotated BRIEF keypoint detection with ECC (ORB + ECC)—selecting the specific method that yields the highest NCC for each individual image file to ensure maximum robustness against local structural deformations (Bethge et al., 2024).

However, a single global transformation matrix frequently fails at the image extremities due to lingering non-linear optical path misalignments, lens distortions, and geometric scanning artifacts inherent to push-broom line scanners (Bethge et al., 2024). To eliminate these peripheral errors, a second-stage fine registration is executed (Bethge et al. 2024). By segmenting the whole canopy into individual plants based on their centroids and performing localized affine transformations on the object-separated sub-images, the spatial overlap ratio (OR_{convex}) is further elevated to $98.0 \pm 2.3\%$ for RGB-to-ChlF and $96.6 \pm 4.2\%$ for HSI-to-ChlF in *Arabidopsis thaliana* datasets, and to $98.9 \pm 0.5\%$ for RGB-to-ChlF and $98.3 \pm 1.3\%$ for HSI-to-ChlF in *Rosa* $\times$ *hybrida* leaf disc datasets (Bethge et al., 2024).

To prevent temporal alignment errors under dynamic field environments, automated multi-sensor gantries require sub-millisecond physical coordination. The TraitDiscover platform achieves this spatiotemporal synchronization by utilizing microsecond-level hardware timestamps and a canonical data schema to co-align five distinct sensor modalities (RGB, 3D LiDAR, thermal, hyperspectral, and chlorophyll fluorescence) (Xu et al., 2026). This precise temporal alignment eliminates leaf-movement and atmospheric artifacts during sequential multi-sensor acquisitions, delivering high-fidelity, co-registered time-series datasets that can detect physiological anomalies—such as water deficit and herbicide injury—days before visible symptoms occur (Xu et al., 2026). This data alignment mechanism operates as a distinct spatiotemporal registration protocol separate from the microsecond-scale electronic hardware-trigger routing used for reactive spray nozzle actuation (Farooque et al., 2023).

Vascular, optical, and chemical co-integration

While optical, thermal, and structural sensor fusion significantly enhances pre-symptomatic diagnostic capabilities, the ultimate frontier of Pre-Symptomatic Crop Intelligence (PSCI) relies on integrating these non-contact canopy sensors with direct vascular and chemical microenvironmental pathways (Garg et al., 2025; Najdenovska, Dutoit, Tran, Plummer, et al., 2021). This vascular-optical-chemical co-integration establishes a multi-dimensional confidence gate, ensuring that early-stage preventative interventions are triggered only when internal systemic defenses and external canopy anomalies align (Garg et al., 2025; Nguyen et al., 2023).

This synergistic integration is exemplified by the AdapTree framework proposed by (Garg et al., 2025), which couples in-vivo plant electrical impedance spectroscopy (EIS) with environmental microclimate sensors (relative humidity and air temperature) in a unified, data-driven architecture. By executing novel four-point-probe electrical impedance measurements directly within the conducting vascular bundles of the plant stem, the system continuously monitors frequency-dependent phase and impedance magnitude variations across a frequency range of 50 Hz to 2 MHz. These electrical metrics directly reflect cell membrane integrity, ion transport, and intracellular water content under dynamic physiological states (García-Vera et al., 2024).

When these internal vascular biosignals are fused with environmental covariates via a boosting-based ensemble model (AdapTree combining AdaBoost and decision trees), the framework achieves an R^2 prediction score of 0.993 for impedance magnitude, and 0.999 for both relative humidity and temperature (Garg et al., 2025). This high predictive accuracy ensures that the plant's internal physiological response is continuously calibrated against microclimatic stressors, preventing environmental fluctuations from inducing false pre-symptomatic stress classifications (Garg et al., 2025).

By linking these contact vascular measurements with non-contact canopy-scale parameters (such as the Photosystem II chlorophyll fluorescence maximum quantum yield, F_v/F_m), the PSCI closed-loop system can cross-verify internal plant-defense activation with external light-harvesting down-regulation (Garg et al., 2025; Xu et al., 2026). This multi-pathway validation elevates detection confidence (C_d) to the near-perfect levels required to justify the immediate execution of targeted variable-rate interventions, effectively closing the loop from sub-visual precursor to precise, site-specific crop protection (Maimaitijiang et al., 2017; Xu et al., 2026).

Edge inference under field constraints

The implementation of pre-symptomatic crop protection is fundamentally constrained by the computational limits of edge hardware and the extreme environmental noise of open-field conditions (Njoroge et al., 2025). Transitioning pre-symptomatic crop intelligence from a theoretical framework to an active field-level intervention requires deploying highly optimized machine learning models on resource-constrained devices, such as

microcontrollers and single-board computers, while maintaining robust, calibrated, and explainable decision boundaries (Gookyi et al., 2024).

From classification to onset forecasting

Standard edge-deployed models in the literature are primarily designed as static, reactive classifiers that identify existing visual symptoms (Malche et al., 2026). While these models are valuable for automating visual inspection, they are fundamentally reactive because their targets are visible symptoms. Pre-symptomatic crop intelligence requires a paradigm shift from static classification to **onset forecasting**—predicting the temporal trajectory of stress progression and the precise "time-to-symptom" window before physical lesions commit to the leaf surface.

This temporal dimension is crucial for extending physiological lead time (L_t) and allowing sufficient actionability. Although historical pest and disease forecasting has relied on coarse macroclimatic models—such as the ARIMAX weather-based coconut rugose spiraling whitefly (RSW) population dynamic forecasting developed by (Elango et al., n.d.)—Pre-symptomatic crop intelligence requires running these forecasting algorithms on-device at the microclimatic scale. Standalone edge classifiers evaluated in the literature—such as Tiny-LiteNet (Nyakuri et al., 2025), TinyLeafNet-IoT (Malche et al., 2026), and MobileNetV3-Small (A. T. Khan et al., 2023)—are cited here specifically for their pioneering model compression, hardware optimization, and edge-scaling architectures, which future pre-symptomatic forecasting systems must inherit to operate under strict field constraints.

Deploying deep neural architectures on highly constrained microcontroller units (MCUs) with limited memory (typically ≤ 50 MB of flash storage and < 1 MB of RAM) represents a significant technical barrier (Gookyi et al., 2024; Njoroge et al., 2025). To bypass these hardware limitations, systematic model compression pipelines must be implemented. For instance, (Gookyi et al., 2024) developed a custom convolutional neural network (CNN) for maize disease detection and deployed it as an optimized 8-bit quantized INT8 model on an Arduino Nano 33 BLE Sense (governed by an ARM Cortex-M4F processor running at 64 MHz). While the unoptimized FLOAT32 model footprint exceeded the 1 MB flash limit of the hardware, post-training INT8 quantization successfully compressed the on-device RAM usage to 726.60 KB and the classifier Flash memory to 344.70 KB, executing a single-image inference in only 11 ms (with a total loop latency of 76.48 ms) and achieving an on-device accuracy of 94.84% (Gookyi et al., 2024).

Similarly, (Malche et al., 2026) designed "TinyLeafNet-IoT," a highly compressed CNN utilizing a MobileNetV2 backbone with a width multiplier of $\alpha = 0.35$ and a reduced input resolution of 96×96 pixels. By applying post-training quantization, the model was compressed to fit within 2.3 MB of Flash memory and 1.1 MB of RAM, achieving a test accuracy of 94.60% and a macro F1-score of 0.94 across five tomato foliar classes (Malche et al., 2026). Deployed on an ESP32-S3 microcontroller operating at 240 MHz, the compressed model achieved an on-device inference latency of under 15 ms per image (compared to a 500 ms unoptimized baseline), consuming less than 140 mW of power and maintaining $95.2\% \pm 2.1\%$ data integrity over a seven-day continuous open-field deployment streaming via MQTT v3.1.1 (Malche et al., 2026).

Beyond manual compression, knowledge distillation offers a powerful mechanism to transfer the representational capacity of high-dimensional ensemble networks into lightweight student architectures (Hasan et al., 2025).

(Hasan et al., 2025) constructed a four-model soft-voting ensemble (DenseNet-121, ResNet-101, DenseNet-201, and EfficientNet-B4) that achieved a peak accuracy of 99.15% on a unified tomato leaf disease dataset. By utilizing temperature-scaled softmax outputs ($T = 15$) and a balanced loss function ($\alpha = 0.7, \beta = 0.3$), this ensemble's dark knowledge was distilled into a compact ShuffleNetV2 student model (Hasan et al., 2025).

The distilled ShuffleNetV2 student retained 98.53% accuracy while achieving a $67.5\times$ parameter reduction (from 91.77M parameters in the ensemble to 1.36M in the student) and a $43.6\times$ execution speedup (Hasan et al., 2025).

Post-training INT8 quantization via the ONNX framework further compressed the student model from 6.00 MB to a mere 1.46 MB (a $671\times$ storage compression relative to the original ensemble), sustaining an accuracy of 97.46% (representing a minor 1.07 percentage point accuracy drop) and enabling an ultra-low inference latency of 0.29 ms on mobile devices (Hasan et al., 2025).

At the physical hardware level, software-hardware co-design and operator-level optimizations are critical to maximize edge throughput. (Lv et al., 2025) optimized a YOLOX-Tiny object detector on an NVIDIA Jetson Nano B01 GPU for peanut foliar disease monitoring. Utilizing the NVIDIA TensorRT framework, they implemented advanced operator fusion, which merges sequential layers into a single execution kernel. From our perspective, such latency gains would be highly critical in pre-symptomatic networks to support high-throughput, multi-channel sampling of fleeting slow wave potentials (SWPs) and transmembrane ion-flux transients without hardware-induced signal aliasing. Specifically, fusing ReLU activation functions with preceding convolutions reduced network execution latency by 55.5% (Lv et al., 2025).

To ensure the fastest and most accurate detection within the TensorRT framework, a custom EfficientNMS-TRT plugin was integrated directly into the ONNX model and compiled (Lv et al., 2025). This optimized GPU non-maximum suppression (NMS) operator runs on CUDA kernels and parallelizes the decoding of bounding boxes, significantly reducing the post-processing bottlenecks (Lv et al., 2025). We extrapolate that accelerating multi-leaf NMS would be essential for real-time robotic scanning platforms to quickly parse hundreds of overlapping leaves per second for dense, pre-visual stress lesions before spatial occlusion errors accumulate. Finally, converting the network tensor layout from the standard NCHW format to the edge-optimized NHWC layout compressed model conversion times by 88.7% and lowered inference latency by 32.3%, enabling a real-time DeepStream video processing throughput of 28.7 FPS at a high resolution of 1280×720 pixels using FP16 precision (Lv et al., 2025).

To ensure practical viability in developing agricultural regions, these optimized edge architectures must be adapted to local socioeconomic constraints (Ibegbu & Korsah, 2024).

In rural West Africa, smallholder farmers operate under severe infrastructural deficits, including a lack of grid electricity, absent cellular connectivity, and extremely low technological literacy. To address these barriers, (Ibegbu & Korsah, 2024) designed a fully offline, battery-powered TinyML diagnostic node utilizing an ultra-low-cost ESP32-CAM AI-Thinker board. Built at a total hardware cost of GHS 280 (less than \$25 USD), this system performs all image acquisition, preprocessing, and model inference locally on-device, completely bypassing the requirement for internet connectivity and mobile data plans, which can otherwise consume 5% to 10% of a rural farmer's monthly household income (Ibegbu & Korsah, 2024; Njoroge et al., 2025).

Similarly, (Njoroge et al., 2025) deployed an edge-optimized hybrid model (combining EfficientNetV2, MobileNetV2, and Vision Transformers) as a 30.4 MB mobile application with a 0.094-second latency. By integrating local environmental sensor readings (soil moisture, DHT11 temperature, and air humidity) directly on-device, this hybrid system reduced false-positive visual diagnoses by 92% through microclimatic correlation without relying on external cloud APIs, validating the potential of fully decoupled, low-cost edge platforms for sustainable smart farming (Njoroge et al., 2025).

Trustworthiness: explainability and uncertainty

In pre-symptomatic crop protection, where interventions are executed prior to any visible symptoms, model trustworthiness is a critical requirement. Triggering site-specific chemical sprayers or micro-irrigation lines based on false pre-symptomatic alarms incurs unnecessary input costs and risks localized environmental toxicity. Conversely, failing to detect an impending stress event allows irreversible damage to proceed. Consequently, on-device models must not only achieve high accuracy but must also be mathematically calibrated to output reliable confidence scores that reflect their actual empirical correctness (Xiang et al., 2026).

The critical necessity of confidence calibration under field conditions is demonstrated by the "overconfident-collapse" phenomenon quantified by (Xiang et al., 2026).

In their cross-domain benchmarking study, a ResNet-50 trained to near-ceiling accuracy (99.73%) under controlled laboratory conditions suffered a catastrophic 67.7-percentage-point accuracy collapse, dropping to only 32.05% when transferred to the heterogeneous field photographs of the PlantDoc dataset. Crucially, despite this near-random accuracy, the model remained unsafely overconfident, outputting a mean predicted confidence of 79.76% across the field test set (Xiang et al., 2026).

This combination of near-random correctness and high confidence is the defining risk of the controlled-to-field reliability gap. This catastrophic collapse is driven by spurious feature exploitation at the pixel level. Image-

level visual statistics confirm large-effect-size shifts between the laboratory and field domains, particularly in HSV saturation ($d = 3.90$), border edge density ($d = 3.33$), and foreground-occupancy proxy ($d = 2.48$) (Xiang et al., 2026).

Grad-CAM visualizations reveal the direct consequences of these shifts: as models transfer to the field, their attention heatmaps drift from central, lesion-centered diagnostic regions to peripheral background textures and border-adjacent areas, elevating the border activation ratio from 0.15 to 0.55 (Xiang et al., 2026).

Standard post-hoc mitigation techniques are insufficient to close this reliability gap. Applying post-hoc temperature scaling calibrated on a grouped field calibration set (Setting B, $T_B = 1.54$) successfully reduced the expected calibration error (ECE) of the fine-tuned ResNet-50 from 0.4761 to 0.3645 (Xiang et al., 2026). However, this calibration improvement is deceptive; even after temperature scaling, the residual ECE remained extremely high, indicating that the model's confidence still significantly exceeded its actual correctness. Selective prediction—allowing the model to abstain and route ambiguous inputs to human experts—is also severely compromised by this calibration gap. At a nominal 80% coverage target, the selective risk of the calibrated ResNet-50 remained unacceptably high at 0.6430 (64.30% error rate among answered samples) because the overconfident background activations bypassed the confidence thresholds (Xiang et al., 2026).

To protect edge devices against these spurious activations and ensure that pre-symptomatic interventions are triggered by authentic physiological indicators, model explainability must be integrated directly into the inference pipeline (Hasan et al., 2025). On mobile platforms, (Hasan et al., 2025) Deployed Grad-CAM++ and Local Interpretable Model-agnostic Explanations (LIME) to provide real-time visual overlays of the model's decision-making process. By generating pixel-level attention heatmaps and feature-importance boundaries, these on-device explainability modules allow agronomists to immediately verify that the distilled student model is focusing on biologically coherent pathological precursors (such as localized chlorotic zones or early necrotic leaf margins) rather than background soil, weeds, or greenhouse gantry structures.

For resource-constrained environments where complex spatial heatmaps cannot be rendered on low-cost screens, (Nyakuri et al., 2025) Integrated a rule-based on-device explainability engine on a Raspberry Pi 5. By mapping the classified Tiny-LiteNet outputs (98.60% accuracy, 16 ms edge latency) directly to local text-based lookups, the system displays natural language descriptions of the predicted pathogen's life cycle, its anticipated agronomic effects, and recommended site-specific integrated pest management (IPM) strategies, converting a black-box edge prediction into a highly interpretable and trust-aware decision-support tool for smallholder farmers (Nyakuri et al., 2025).

As established by (Xiang et al., 2026) These standard calibration, selective prediction, and post-hoc mitigation limits underscore a profound research gap. The fact that standard selective prediction risk remains above 0.51 even when paired with advanced transformers and deep ensembles demonstrates that post-hoc scaling alone cannot solve the controlled-to-field shift. This persistent reliability gap underpins the urgent research needs and future directions of the field, setting up a critical transition to our upcoming analysis of domain adaptation and perennial field deployment.

Closing the loop: decision, actuation, and feedback

The transition from reactive, symptom-triggered intervention to anticipatory crop protection requires closing the loop between pre-symptomatic detection and site-specific variable-rate actuation. Traditional crop protection models are predominantly symptom-driven, initiating chemical application or mechanical disturbance only after visible damage has been committed and the available intervention window has significantly narrowed. While closed detect-to-spray loops have been successfully implemented for visible targets—such as the integrated UAV-deep learning and pulse-width modulation (PWM) spraying systems reviewed by (W. Li et al., 2025), or the real-time machine vision-based targeted weed control architectures developed by (Liu et al., 2021) and evaluated in the field by (Farooque et al., 2023)—These systems remain fundamentally reactive because their triggering signals are coupled to physical, symptomatic tissue damage. In contrast, pre-symptomatic crop intelligence establishes a closed loop before visible damage by mapping pre-visual physiological, biochemical, or electrophysiological changes directly to predictive actuation commands, preventing the stress cascade from progressing to symptomatic expression.

Decision support and actuation latency

Actuation latency represents a critical physical bottleneck in translating a pre-symptomatic decision into a localized, precise field intervention. In targeted variable-rate spraying, the physical response time of the system encompasses the delay between sensor-based target identification, computational decision generation, controller signal routing, and the physical opening of the nozzle (Jiao et al., 2025).

For instance, (Liu et al., 2021) developed an electronic targeted spraying system utilizing high-response 12 V solenoid valves with a physical response time of less than 60 ms to actuate individual nozzles spaced 0.7 m apart in a strawberry crop. To minimize the computational delay of deep learning models, (Farooque et al., 2023) utilized parallel processing on graphical processing units (GPUs) to compress the inference and triggering latency, enabling the co-location of cameras and spray nozzles at the rear of a vehicle operating at speeds of up to 2.77 m/s. While the edge-GPU frame inference latency operates at the millisecond scale (approximately 32 ms), parallelized thread execution and high-frequency serial signals compress downstream electronic hardware trigger routing between the processing unit and the nozzle relays to the microsecond level (Farooque et al., 2023).

In orchard and perennial canopy systems, actuation complexity is heightened by the three-dimensional morphology of the crop, requiring geometric adjustments to ensure optimal deposition (Nan et al., 2023). To address this spatial variability, (Nan et al., 2023) developed a profile variable sprayer that automatically adjusts its mechanical arms to fit the contour of the tree canopy. By utilizing an 8-channel ultrasonic sensor array to measure canopy distance and calculating target profile angles via an improved cerebellar model articulation controller (CMAC-PID) algorithm, the overall mean tracking error was reduced by 35.0% compared to traditional controllers, allowing on-demand pesticide application tailored to tree height, volume, and leaf area index (Nan et al., 2023).

The specific variable-rate spray architecture deployed—whether pressure-regulated, flow-rate-regulated, or pesticide concentration-regulated—fundamentally defines the system’s actionability and response latency (Jiao et al., 2025).

Pressure-regulated architectures adjust flow rates by modulating system pressure, but they are limited by a narrow regulation range, slow response times (e.g., 1.5 to 2.35 s of total delay), and undesirable pressure-induced changes in droplet size distribution (Jiao et al., 2025).

Flow rate-regulated systems utilizing PWM solenoid valves at the nozzle bodies offer significantly faster response times (typically 10 to 50 ms) and decouple flow rate from spray pressure, yet they suffer from mechanical wear of the valve spool, pipeline pressure fluctuations, and nonlinear flow-to-duty-cycle relationships at extreme duty cycles, particularly below 20% (Jiao et al., 2025; W. Li et al., 2025).

Pesticide concentration-regulated systems utilizing direct injection or inline jet mixing avoid chemical residues in the main water tank but introduce systemic transport mixing delays and remain highly sensitive to pesticide fluid viscosity, requiring real-time viscosity-compensating control strategies (Jiao et al., 2025).

Actuation in a pre-symptomatic loop must dynamically integrate these mechanical constraints to suppress upstream positioning errors, which can otherwise range from 0.3 to 0.8 m under dynamic field conditions (W. Li et al., 2025).

When properly calibrated, these variable-rate actuation technologies achieve pesticide reduction rates of 30% to 50% and off-target drift reductions of more than 30% without sacrificing crop protection efficacy (W. Li et al., 2025; Zanin et al., 2022).

The predictive field-state node (a constrained digital twin)

Closing the pre-symptomatic loop under operational field conditions requires a virtual representation capable of simulating unobserved system dynamics, quantifying predictive uncertainty, and continuously assimilating real-time observations. This virtual layer is established through a predictive field-state model, which functions as a constrained digital twin of the crop-environment system. Unlike unconstrained data-driven models that exhibit high sensitivity to environmental noise and suffer from accuracy collapse when transferred from laboratory to field environments, a constrained digital twin marries mechanistic, physics-based crop models with data-driven machine learning algorithms to constrain the search space of the (Kallenberg et al., 2025).

This cyber-physical integration is demonstrated by the deployment of interoperable agricultural digital twins that couple process-based crop growth models (such as WOFOST) and plant-pathogen models (such as A-scab) with offline-trained reinforcement learning (RL) agents (Kallenberg et al., 2025). In this architecture, the digital twin is maintained in synchrony with the physical field through continuous data assimilation, utilizing a variational recalibration approach (Kallenberg et al., 2025). Ground-based measurements—including Leaf Area Index (LAI) and phenological development stages (BBCH scale)—are periodically captured and transmitted via low-latency IoT protocols (such as NGSI-LD) to a centralized Orion Context Broker (Kallenberg et al., 2025). The digital twin automatically executes optimization algorithms (such as the Subplex algorithm) to recalibrate core, non-differentiable model parameters within physiologically plausible bounds, minimizing the discrepancy between simulated trajectories and real-world observations (Kallenberg et al., 2025).

Once synchronized, the predictive field-state node serves as a "what-if" simulator to evaluate the long-term agronomic and environmental consequences of proposed interventions before they are physically executed (Awais et al., 2025; Escribà-Gelonch et al., 2024; Melesse, 2025).

Rather than relying on rigid, rule-based thresholds, the offline-trained RL agent receives the current multi-dimensional state vector from the digital twin (e.g., soil inorganic nitrogen, crop phenology, and forecasted temperature sum trajectories) and outputs an optimized recommendation, such as a specific nitrogen dose or a preventative fungicide volume (Kallenberg et al., 2025).

The digital twin then simulates the prospective crop development and disease risk over a forecast window, confirming that the recommended action maximizes resource-use efficiency and avoids environmental leaching or pathogen infection before transmitting the command to physical actuators (Kallenberg et al., 2025).

The subsequent physiological feedback of the crop to the applied treatment (e.g., changes in transpiration rate, electrical potential, or chlorophyll fluorescence) is recorded by the sensor array and fed back to the virtual model, completing the closed-loop "sensing-inference-decision-actuation-feedback" paradigm (Awais et al., 2025; Peladarinos et al., 2023).

Deployment in resource-constrained and perennial systems

The physical translation of Pre-Symptomatic Crop Intelligence (PSCI) from idealized laboratory environments to heterogeneous field conditions represents the primary operational bottleneck in precision plant protection. While laboratory evaluations consistently yield diagnostic accuracies exceeding 95%, the transition to open-canopy environments exposes severe performance-adoption challenges (Shafay et al., 2025). This section quantifies the cross-domain reliability gap, instantiates the closed-loop architecture through a flagship perennial pest case study, and evaluates the unique economic and computational constraints of deploying pre-symptomatic systems within perennial orchards and smallholder farming environments.

The field-versus-laboratory gap

The discrepancy between laboratory diagnostic performance and field-deployment accuracy represents a critical "reliability gap" that directly threatens the operational viability of pre-symptomatic interventions (Xiang et al., 2026). When deep learning architectures trained on standardized laboratory datasets—such as PlantVillage, which features isolated, single leaves photographed against uniform backgrounds under controlled lighting—are deployed in real-world field settings, their classification performance collapses dramatically (Ahmad et al., 2023; Moupojou et al., 2023).

This performance degradation is driven by severe environmental domain shifts, including dynamic solar illumination, shadow casting, wind-driven leaf movement, camera-viewing angle variations, and the presence of complex agricultural backgrounds containing soil, mulch, weeds, and overlapping foliar structures (Arsenovic et al., 2019; Shafay et al., 2025).

For instance, in a comprehensive benchmarking study, standard convolutional neural network (CNN) architectures trained on controlled images exhibited a severe accuracy collapse when tested on natural, field-acquired images (Shafay et al., 2025).

ResNet50 suffered a catastrophic **56.39 percentage point decline**, dropping from a laboratory validation accuracy of **98.18% to a field-testing accuracy of only 41.79%** on the PlantDoc dataset (Shafay et al., 2025).

Similarly, the modern ConvNeXt architecture declined by **27.71 percentage points**, collapsing from **99.47% on PlantVillage to 71.76% on PlantDoc** (Shafay et al., 2025).

Under the same field transfer, standard models like AlexNet, VGG19, InceptionV3, and ResNet152 trained strictly under controlled laboratory conditions struggled to exceed **81% to 84% accuracy** when confronted with field-collected leaves surrounded by complex backgrounds (Arsenovic et al., 2019).

In cross-domain few-shot learning (CD-FSL) tasks, where models must generalize to previously unseen disease categories and entirely new visual environments, the domain gap becomes even more pronounced (S. Yang et al., 2024). Standard few-shot classifiers that perform with near-perfect accuracy under single-domain configurations exhibit sharp performance drops when evaluated on field crop disease datasets (S. Yang et al., 2024). When transferring feature representations across a high domain gap (e.g., from an insect pest source domain to a foliar crop disease target domain), advanced metric-based models like CDFSL-BDC achieve accuracies of only **63.95% in the 1-shot setting and 80.13% in the 5-shot setting** (S. Yang et al., 2024).

This extensive field degradation highlights why establishing calibrated detection confidence (C_d) is a fundamental prerequisite for closed-loop interventions. Standalone edge classifiers are frequently highly overconfident despite their collapsing accuracy under field conditions, with some models maintaining a max probability of **79.76% while actual field accuracy collapses to 32.05%** (Xiang et al., 2026). This overconfidence triggers highly destructive false-alarm cascades; if a pre-symptomatic warning is triggered with high nominal confidence but low actual reliability, growers are forced to deploy expensive, preventative variable-rate spray treatments or anti-transpirants on healthy plants, leading to wasted inputs, increased operational costs, and phytotoxicity risks (Shafay et al., 2025; Xiang et al., 2026).

To mitigate this gap, researchers have developed robust dataset architectures that capture natural field variability directly at the source (Moupojou et al., 2023). The FieldPlant dataset, which includes **5,170 plant disease images** collected directly from Cameroonian plantations under diverse natural illumination and complex backgrounds, provides a highly realistic field benchmark containing **8,629 annotated, overlapping leaves** across 27 disease classes (Moupojou et al., 2023; Shafay et al., 2025). Under supervised evaluations, classifiers trained on FieldPlant's realistic conditions consistently outperform those trained on internet-scraped datasets (such as PlantDoc) due to the complete exclusion of laboratory-style images and the precise, pathologist-supervised annotation of natural foliar clusters (Moupojou et al., 2023).

Simultaneously, targeted data-augmentation and preprocessing techniques have been developed to bypass background-induced noise (Ahmad et al., 2023). By introducing an alpha channel to generate red, green, blue, and alpha (RGBA) images and setting the background pixel values to zero, researchers can train models that ignore the surrounding soil and weed artifacts, forcing the network to focus exclusively on lesion-level physiological characteristics (Ahmad et al., 2023). Utilizing this RGBA background-removal technique on the custom Corn Disease & Severity (CD&S) dataset, the DenseNet169 architecture achieved a peak cross-dataset generalization accuracy of **81.60%** under diverse field conditions (Ahmad et al., 2023). Furthermore, augmenting controlled laboratory datasets with even a **10% fraction of field-acquired images** has been shown to raise cross-dataset generalization accuracies from **45.36% to 79.50% on PlantVillage** and from **61.00% to 79.50% on PlantDoc**, demonstrating that training data diversity is a powerful lever for overcoming the field-laboratory reliability gap (Ahmad et al., 2023).

Three recurring cases: RSW on coconut, water stress, and nitrogen deficiency

To demonstrate the real-world utility and biological generalizability of the PSCI closed-loop framework, we instantiate the "sensing–inference–decision–actuation–feedback" cycle across three recurring stress case studies: a principal biotic pest system (Rugose Spiraling Whitefly on coconut), an abiotic soil-water deficit pathway (water stress), and an abiotic nutritional stress pathway (nitrogen deficiency).

Flagship case study: Rugose Spiraling Whitefly on coconut

The Rugose Spiraling Whitefly (RSW, *Aleurodicus rugioperculatus* Martin) represents a devastating, exotic invasive sap-sucking pest that infests tropical perennial coconut palms (*Cocos nucifera*) across coastal India (Elango & Nelson, 2020; Sundararaj & Selvaraj, 2017). The biological progression of RSW infestation represents a classic multi-stage stress cascade (Elango & Nelson, 2020). First-stage nymph feeding directly drains sap, removing critical water and nutrients from the vascular system of the bottom matured fronds (Elango & Nelson, 2020). As the infestation advances, the nymphs excrete immense volumes of sticky, sugar-rich honeydew, which coats the lower fronds and provides an ideal substrate for the rapid proliferation of black sooty mould (Elango & Nelson, 2020). Although the sooty mould itself is non-pathogenic and does not infect the plant tissue, its thick, opaque physical layer blocks solar radiation, severely reducing the local leaf transpiration, stomatal conductance, and Photosystem II (PSII) photosynthetic efficiency, eventually triggering severe foliar chlorosis, necrosis, and premature frond drop (Elango & Nelson, 2020; Sundararaj & Selvaraj, 2017).

By instantiating the PSCI closed-loop framework, this stress cascade can be suppressed pre-symptomatically—long before sooty mould or chlorosis develops—through a coordinated sensing, inference, and actuation cycle (Fig. 4):

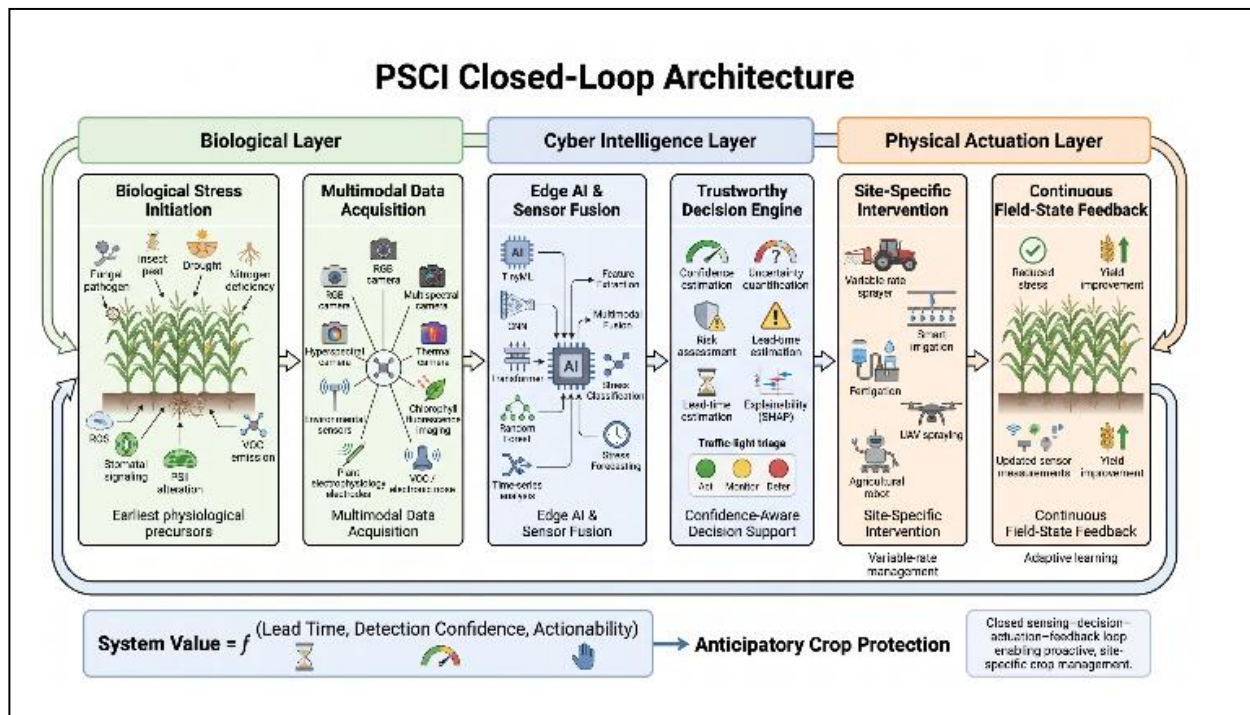


Fig. 4 Instantiation of the PSCI closed loop for Rugose Spiraling Whitefly (*Aleurodicus rugioperculatus*) on coconut, tracing the sensing, inference, decision, actuation, and feedback stages.

1. **Sensing:** The earliest physiological precursors of RSW feeding are captured pre-symptomatically before visual foliar greenness changes. Electrophysiological variation potentials and transmembrane ion-flux transients are expected to depolarize rapidly within conducting vascular bundles in response to nymph sap-extraction, as established for analogous sap-sucking pests (Najdenovska, Dutoit, Tran, Plummer, et al., 2021). Concurrently, leaf-level chlorophyll fluorescence imaging captures early Photosystem II perturbations, manifested as dark-adapted non-photochemical quenching (NPQ) and potential activity (F_v/F_0) deviations, by analogy to ChIF nematode-stress detection (van Himbeek et al., 2025). At the field scale, microclimatic sensors record ambient weather covariates, as RSW population buildup is heavily dependent on seasonal weather patterns (Elango & Nelson, 2020).
2. **Inference and Onset Forecasting:** Raw multi-channel electrophysiological time-series and non-contact canopy reflectance parameters are processed at the edge using lightweight machine learning classifiers (such as Random Forests or SVMs) to detect sub-visual anomalies (H. Li et al., 2024; Macdougall et al., 2022).

To resolve the environmental noise of open-canopy sensing, these physiological features are fused with meteorological data and parsed through temporal forecasting networks (Elango et al., n.d.).

Building upon the historical ARIMAX time-series forecasting baseline—where an ARIMA(0,2,1) model fitted with maximum temperature as the sole significant weather covariate successfully predicted the weekly surge of RSW nymphs per leaf—the edge-inference system projects the future trajectory of the insect population (Elango et al., n.d.).

This forecasting step allows the system to predict when nymph densities will escalate from their post-rainy minimum (July: 130.8 nymphs/leaf/frond) toward their summer epidemic peak (October: 161.0 nymphs/leaf/frond), providing a critical action window before irreversible visual damage is committed (Elango et al., n.d.).

3. **Decision and Actuation Gating:** Once the forecasted population trajectory crosses the actionability threshold, the decision support system evaluates the optimal site-specific intervention (Kallenberg et al., 2025).

The response latency of the intervention defines its actionability. In smallholder contexts, immediate chemical spraying is often restricted by high input costs, water scarcity, and environmental toxicity, while prolonged dry spells and deficit monsoons act as predisposing weather factors that accelerate pest proliferation (Elango & Nelson, 2020).

Under the PSCI framework, the decision support system coordinates a biological-first actuation strategy. Because the percent parasitism of the native parasitoid *Encarsia* spp. is highly negatively correlated with the RSW nymph population ($r = -0.684^{**}$), the system calculates the required release volume of *Encarsia guadeloupae* or *Chrysoperla zastrowi sillemi* to suppress the upcoming pest surge naturally (Elango and Nelson 2020).

If chemical or biological spot-spraying is deemed necessary, the actuation commands are routed directly to variable-rate sprayers, restricting application strictly to the bottom matured five fronds of the target palms—which represent the primary niche of nymph colonization—minimizing off-target ground deposition compared to continuous sprayers (Elango & Nelson, 2020).

4. **Loop Closure and Feedback:** Following parasitoid release or targeted variable-rate spraying, the loop is closed by continuous physiological feedback monitoring. The reduction in nymph feeding pressure is captured as stem-impedance phase angles stabilize and light-adapted Photosystem II fluorescence parameters (Φ_{PSII}) return to healthy baselines (Garg et al., 2025; Lee et al., 2025). At the canopy scale, the recovery of transpiration-driven evaporative cooling is tracked via thermal imaging, while the suppression of secondary sooty mould development is verified via NIR reflectance, directly informing the predictive field-state digital twin to refine the next forecasting iteration (Chaerle et al., 2004; Kallenberg et al., 2025).

Supporting abiotic cases: water stress and nitrogen deficiency

To prove the generalizability of this closed-loop architecture, the same "sensing–inference–decision–actuation–feedback" workflow is traversed under abiotic stress conditions:

- **Water Stress:** Soil water deficit (drought) is sensed pre-symptomatically as early as day 4—well before visible foliar wilting or greenness changes occur—via stomatal-conductance temperature increases captured by thermal imaging, combined with light-adapted chlorophyll fluorescence perturbations (NPQ and Φ_{PSII} deviations) (Chaerle et al., 2004; Xu et al., 2026). The edge-inference system fuses these thermodynamic and photochemical features to estimate transpiration deficits under dynamic microclimates (Zandi et al., 2025). The decision support system resolves the water-deficit trajectory, automatically generating a variable-rate irrigation prescription that triggers localized solenoid drip valves or pulse-irrigation actuators, directly restoring soil moisture in high-deficit zones (Jiao et al., 2025; Nackley et al., 2021). Closed-loop feedback is established as leaf temperature drops from transpiration-driven evaporative cooling and Photosystem II vitalities (R_{fd}) recover, updating the soil-canopy digital twin (Chaerle et al., 2004; Kallenberg et al., 2025; Park et al., 2024).

- **Nitrogen Deficiency:** Plant nutritional stress is captured pre-symptomatically 1 to 2 days before visual foliar chlorosis develops by executing high-impedance, multi-channel electrophysiological measurements along the petiole to record systemic variation potential and ion-flux transients, combined with narrow-band red-edge HSI spectral indices (García-Vera et al., 2024; González I Juclà et al., 2023).

Supervised machine learning classifiers trained on on-device electrophysiological signatures isolate the nitrogen-deficit profile from confounding water-deficit anomalies (González I Juclà et al., 2023).

The decision support system generates a zone-specific variable-rate nitrogen fertilization map, directing PWM-controlled liquid fertilizer actuators to apply precise nitrogen doses only to deficient canopy zones, preventing environmental leaching and reducing chemical waste (Jiao et al., 2025; W. Li et al., 2025).

Feedback is closed as transmembrane potentials repolarize and foliar red-edge reflectance shifts back to healthy baselines, confirming cellular nitrogen assimilation (García-Vera et al., 2024; González I Juclà et al., 2023).

Perennial economics and smallholder constraints

The deployment of pre-symptomatic crop protection systems faces distinct horticultural, economic, and computational constraints in perennial orchards and smallholder farming environments compared to annual row crops (Nackley et al., 2021; Shafay et al., 2025).

Orchard variable-rate spraying and canopy dynamics

Deciduous perennial crops—such as apple orchards (*Malus domestica*) and grape vineyards (*Vitis vinifera*)—exhibit dynamic, three-dimensional canopy architectures that undergo extreme changes in volume, density, and leaf area index across seasonal phenophases (BBCH scale) (Nackley et al., 2021). Standard constant-rate air-assisted sprayers discharge a uniform volume of pesticide regardless of these structural shifts, resulting in massive chemical waste and environmental drift during early-season sparse phenophases, and severe under-spraying during full-canopy mature phenophases (Nackley et al., 2021). Because fruit trees and grapevines represent significant long-term capital investments with multi-year lifespans, a single unchecked disease outbreak can permanently damage the woody architecture, causing catastrophic cumulative yield losses over multiple seasons (Nackley et al., 2021; Shafay et al., 2025). Consequently, the economic payback of early-stage, pre-symptomatic detection is substantially higher in perennial crops than in annual row crops (Nackley et al., 2021).

To optimize pesticide applications across these seasonal shifts, conventional air-blast sprayers can be retrofitted with laser-guided, computer-controlled variable-rate systems (Nackley et al., 2021). Utilizing canopy-scanning sensors (such as LiDAR and ultrasonic arrays, see the closed-loop actuation analysis) combined with Doppler radar ground speed sensors, these systems dynamically scan the crop structure in real-time, automatically calculating the tree row volume to adjust individual nozzle flow rates via PWM solenoid valves (Nackley et al., 2021). In multi-seasonal field trials, retrofitted variable-rate sprayers demonstrated massive resource-use efficiencies (Nackley et al., 2021):

- **Orchard (Apple):** Applied **67% to 74% less pesticide volume** in the variable-rate mode compared to the constant-rate mode across full bloom, fruit fall, and advanced ripening stages (Nackley et al., 2021).
- **Vineyard (Grape):** Applied **61% to 80% less pesticide volume** across prebloom, full bloom, and full canopy stages (Nackley et al., 2021).
- **Drift and Ground Losses:** Reduced off-target spray losses by **up to 40% in orchards and up to 33% in vineyards** (Nackley et al., 2021).

Furthermore, the variable-rate mode maintained a consistent, targeted canopy deposit density (averaging **70 to 120 droplets/cm²**), successfully preventing "overspray" conditions where droplets coalesce on the leaf surface into large, flowing sheets, causing pesticide runoff into the soil (Nackley et al., 2021). Based on this verified performance, laser-guided variable-rate air-assisted sprayers are formally recognized as Drift Reduction Technologies (DRTs) under United States Environmental Protection Agency (USEPA) guidelines, providing a validated engineering pathway for commercial adoption (Nackley et al., 2021).

UAV variable-rate spraying and computational bottlenecks

To overcome the terrain-limited mobility of ground machinery in hilly, mountainous, or steep-slope orchards, unmanned aerial spraying systems (UASS) have emerged as highly maneuverable variable-rate platforms (Chen et al., 2025). The state-of-the-art UASS workflow integrates high-precision 3D canopy perception with targeted drone actuation (Chen et al., 2025):

1. **3D Canopy Perception:** A remote sensing UAV (such as the DJI Matrice 300 RTK equipped with a Zenmuse L1 LiDAR-RGB fusion gimbal) scans the orchard, generating high-density 3D point clouds (Chen et al., 2025).
2. **Point Cloud Processing:** Ground points are classified using an Improved Progressive TIN Densification (IPTD) filtering algorithm, and individual tree canopies are segmented using a region-growing algorithm applied to the Canopy Height Model (CHM) (Chen et al., 2025). This extracts precise individual-tree parameters, including tree height, crown diameter, and canopy volume (Chen et al., 2025).
3. **Prescription and Actuation:** Canopy volumes are classified into discrete volume-graded spray levels (typically Class I to IV) to generate a high-resolution prescription map (Chen et al., 2025). A plant protection UAV (such as the DJI T30) executes the prescription using a "distributed multi-point spraying strategy," maintaining uniformly distributed application points in the canopy projection plane to maximize droplet penetration and uniformity (Chen et al., 2025).

In extensive field validation trials across citrus (*Citrus reticulata* Blanco) and litchi (*Litchi chinensis* Sonn.) orchards, this UAV variable-rate workflow achieved exceptional performance (Chen et al., 2025). In the litchi orchard, variable spraying on large tree canopies achieved a maximum canopy coverage of **14.47%**, while reducing non-target ground deposition from **20.60 $\mu\text{g}/\text{cm}^2$ to only 1.98 $\mu\text{g}/\text{cm}^2$** —representing a massive **90.4% reduction in pesticide soil loss** compared to traditional continuous spraying (Chen et al., 2025). In the citrus orchard, the variable-rate treatment reached a maximum canopy coverage of **9.68%**, while reducing ground deposition from **5.16 $\mu\text{g}/\text{cm}^2$ to 1.85 $\mu\text{g}/\text{cm}^2$** —representing a **64.1% reduction in pesticide soil loss** (Chen et al., 2025).

Despite these impressive chemical savings, significant **physical and operational bottlenecks** constrain large-scale commercial UASS adoption (Chen et al., 2025):

- **Data-Decision Latency:** The offline processing of 3D point clouds, TIN densification, individual tree segmentation, and prescription map generation takes approximately **20 minutes per hectare** (20 min/ha) on high-performance workstations (Chen et al., 2025). This high latency prevents real-time, on-the-fly "scan-and-spray" operations, requiring separate, pre-planned mapping and spraying flights (Chen et al., 2025).
- **Energy and Time Penalties:** The "stop-and-spray" strategy adopted in UASS variable spraying—where the drone must repeatedly hover over each target tree, trigger a localized spray pulse according to the volume prescription, and then accelerate to the next tree—inevitably leads to **significantly increased battery energy consumption and longer operational times** compared to continuous, constant-speed flight paths (Chen et al., 2025).

Smallholder adoption barriers and context-specific scaling

Deploying pre-symptomatic crop protection in smallholder farming systems across developing agricultural regions (such as Sub-Saharan Africa or Southeast Asia) introduces severe socio-economic and infrastructural barriers that cannot be resolved by high-cost, industrial-scale technologies (Shafay et al., 2025). Widespread adoption is strictly bound by three critical deployment constraints (Shafay et al., 2025):

1. **Economic Barriers (CAPEX):** Research-grade hyperspectral imaging (HSI) systems cost between **\$20,000 and \$50,000**, presenting an absolute financial barrier for smallholders (Shafay et al., 2025). In contrast, low-cost RGB-based smartphone applications or portable, custom-built multi-sensor IoT nodes costing **under \$25** are highly accessible, aligning directly with local smallholder purchasing power (Ibegbu & Korsah, 2024; Shafay et al., 2025).

2. **Infrastructure and Connectivity Limits:** Rural smallholder regions frequently suffer from unstable power grids and a complete lack of cellular internet connectivity (Shafay et al., 2025). Cloud-dependent disease classification platforms (such as Plantix, which requires active internet to upload photos to external deep learning servers) are unusable in remote fields (Shafay et al., 2025). Practical field solutions must prioritize **offline functionality**, running compressed TinyML classifiers directly on low-cost edge microcontrollers or smartphones to eliminate cellular data dependency (Ibegbu & Korsah, 2024; Shafay et al., 2025).
3. **Expertise and Trust Barriers:** Smallholders lack formal training in spectral signal processing or high-dimensional data analysis (Shafay et al., 2025). Complex "black-box" models that output abstract stress ratings fail to build user trust (Shafay et al., 2025). Successful adoption requires **intuitive, explainable AI (XAI)** decision support—such as built-in text-based local explanations or localized mobile-app recommendations—to align AI-driven pre-symptomatic diagnoses with traditional agricultural expertise and local farming practices (Nyakuri et al., 2025; Shafay et al., 2025). By scaling advanced sensing and inference architectures down to these context-specific constraints—such as deploying offline, smartphone-optimized classifiers like PlantVillage Nuru, which runs deep learning models locally to diagnose cassava diseases with twice the accuracy of local extension workers—the PSCI framework can be democratized, delivering high-leverage, pre-symptomatic crop intelligence to the most vulnerable agricultural communities globally (Shafay et al., 2025).

A unifying framework for pre-symptomatic crop intelligence

The transition from reactive, symptom-triggered plant protection to an anticipatory management paradigm requires the structural integration of advanced sensing, machine learning inference, and variable-rate actuation into a single, cohesive cyber-physical system (Kallenberg et al., 2025; W. Li et al., 2025).

While previous literature has focused heavily on isolated components—such as comparing the classification accuracies of different deep learning models or testing the mechanical performance of spraying nozzles—this fragmentation has obscured the systemic bottlenecks that govern overall field performance (García-Vera et al., 2024; W. Li et al., 2025).

The Pre-Symptomatic Crop Intelligence (PSCI) framework unifies these disparate research fields into a cyber-physical loop, providing a standardized blueprint for timely, confidence-gated crop protection (Kallenberg et al., 2025).

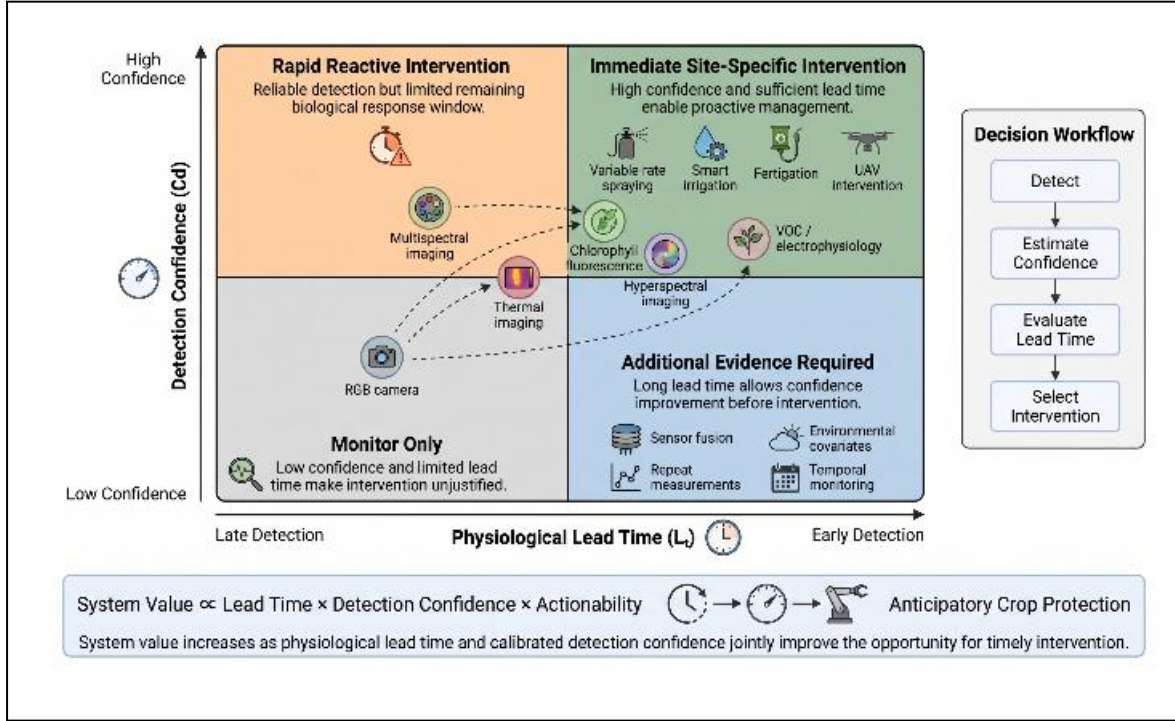
The PSCI reference architecture comprises five interconnected nodes: (1) Sensing, (2) Inference, (3) Decision and Confidence Gating, (4) Actuation, and (5) Feedback and Simulation (Kallenberg et al., 2025).

This unified cyber-physical loop operates as an advanced "health check for plants," converting transient, sub-visual crop stress signals into localized, highly calibrated actions (Kallenberg et al., 2025).

The coupling framework and the thesis of system value

The underlying economic and agronomic value (V) of a pre-symptomatic crop protection system is not determined by its most advanced component, but is strictly bounded by its weakest cyber-physical link (Kallenberg et al., 2025; W. Li et al., 2025). This governing principle is formally expressed through the coupled thesis equation (Kallenberg et al., 2025):

Fig. 5 The PSCI closed-loop reference architecture spanning the biological, cyber-intelligence, and physical-



actuation layers, from sub-visual precursors to variable-rate actuation and predictive field-state feedback.

$$V = f(L_t, C_d, A_c) \quad \text{where} \quad \lim_{\min(L_t, C_d, A_c) \rightarrow 0} V = 0$$

where L_t is the physiological lead time of the stress signal (defined as the duration between the onset of sub-visual physiological stress and the eventual manifestation of visible foliar symptoms) (Xu et al., 2026; Zhu et al., 2017), C_d is the calibrated detection confidence or classification reliability under heterogeneous field shifts (Hasan et al., 2025; Xiang et al., 2026), and A_c is the actuation actionability, defined as the inverse of the response latency (R_l) required to execute the physical intervention in the field (W. Li et al., 2025):

$$A_c = \frac{1}{R_l}$$

This mathematical coupling reveals that a failure in any single node of the loop completely neutralizes the system's agronomic utility (Kallenberg et al., 2025).

For instance, a pre-symptomatic signal that detects vascular stress extremely early ($L_t = 4$ d) (Xu et al., 2026) but with very low reliability ($C_d \rightarrow 0$) (Xiang et al., 2026) cannot justify the financial and environmental costs of preventative chemical or biological treatments, rendering the early lead time useless. Conversely, even if a model identifies stress with absolute confidence ($C_d \rightarrow 1$) and high lead time ($L_t = 4$ d) (Xu et al., 2026), but the mechanical delivery mechanism has a sluggish response latency ($R_l \rightarrow \infty$; $A_c \rightarrow 0$), the physical treatment will be executed too late, missing the pre-symptomatic action window and resulting in irreversible foliar damage (W. Li et al., 2025).

Furthermore, confidence and actionability are tightly coupled; low-confidence alerts cause growers to hesitate or defer, artificially inflating response latency and collapsing the loop value (Xiang et al., 2026). Thus,

designing an effective pre-symptomatic system requires researchers to prioritize "loop symmetry"—ensuring that sensing lead times, inference calibration, and mechanical actuation are balanced and optimized in unison, rather than allowing a single high-performance component to be bottlenecked by weaker nodes (Kallenberg et al., 2025).

The decision-space diagram

To operationalize this coupling under field conditions, the PSCI framework maps the system state onto a two-dimensional decision-space diagram (Fig. 6), partitioning the $L_t \times C_d$ (lead-time $\times$ confidence) plane into three distinct operational regimes:

1. **Regime I: Act Now / Intervene (High L_t , High C_d):** When both physiological lead time and calibrated detection confidence exceed established thresholds ($L_t \times C_d \rightarrow \text{high}$), the system immediately triggers site-specific, variable-rate actions (such as localized micro-dose spraying, nutrient dosing, or targeted biological releases) to suppress the stress vector before visible tissue damage is committed (W. Li et al., 2025).
2. **Regime II: Monitor / Re-evaluate (High L_t , Moderate/Low C_d):** If the lead-time signal is early but the calibrated confidence falls below the selective-prediction threshold ($C_d < t$), the system abstains from immediate, potentially wasteful chemical interventions (Xiang et al., 2026). Instead, it schedules targeted, high-frequency diagnostic scans—such as deploying autonomous gantries or high-throughput UAV flyovers—to track the localized anomaly and accumulate additional spatial-spectral evidence until a confident decision can be rendered (Xu et al., 2026).
3. **Regime III: Defer / No Action (Low L_t , Any C_d):** If the physiological lead time approaches zero ($L_t \rightarrow 0$), the pre-symptomatic window has completely closed. The stress has progressed to visible foliar symptoms (chlorosis, necrosis), and the system defers to conventional, reactive, symptom-stage spot-spraying or standard agricultural management (W. Li et al., 2025).

By managing pre-symptomatic warnings through these risk-graded regimes, the decision support system protects growers from costly false-alarm cascades while maximizing the environmental and economic returns of early-stage, site-specific crop protection (W. Li et al., 2025; Xiang et al., 2026).

Unified mapping of literature streams

The modular architecture of the PSCI closed-loop system successfully unifies the fragmented research papers within the precision agriculture and plant science literature, mapping each separate study to a defined, functional node in the loop (Kallenberg et al., 2025):

- **Sensing Node:** Biological precursor signals are captured through in-vivo transmembrane potential transients and variation potentials recorded outside a Faraday cage (Tran et al., 2019), or via pre-visual near-infrared (NIR) and chlorophyll fluorescence (ChlF) kinetics (Xu et al., 2026; Zhu et al., 2017).
- **Inference Node:** High-dimensional data cubes are compressed using the Successive Projections Algorithm (SPA) for waveband selection (Zhu et al., 2017) and deployed as quantized, lightweight convolutional neural networks running on resource-constrained single-board edge computers (Gookyi et al., 2024; Nyakuri et al., 2025).
- **Decision and Confidence Gating Node:** The severe domain-transfer reliability gap and spurious feature activations are managed through post-hoc temperature scaling, selective-prediction coverage-risk mapping, and Grad-CAM++ biological feature-attention overlays to ensure the model utilizes disease-centered lesions rather than background clutter (Hasan et al., 2025; Xiang et al., 2026).
- **Actuation Node:** Decisions are executed as precise, zone-specific variable-rate applications, utilizing high-speed solenoid valves and real-time inline pesticide mixing systems to reduce chemical volumes by 30% to 50% and off-target drift by over 30% (W. Li et al., 2025; Nan et al., 2023).
- **Feedback and Simulation Node:** Interoperable agricultural digital twins use variational data assimilation and the Subplex algorithm to continuously calibrate crop models like WOFOST and A-scab, allowing

reinforcement learning (RL) policies to simulate "what-if" scenarios and optimize the next iteration of the closed loop (Kallenberg et al., 2025).

Table 3: Technology Readiness Level (TRL) Staging of the PSCI closed loop

To guide future funding and research initiatives, Table 3 establishes a standardized Technology Readiness Level (TRL) staging for each component of the PSCI closed-loop reference architecture (Kallenberg et al., 2025). By defining primary functions, hardware or algorithmic inputs, literature baselines, and engineering bottlenecks, this staging-maps the transition of pre-symptomatic crop intelligence from a laboratory-controlled curiosity to an operational, field-deployed agricultural reality.

Importantly, the inclusion of wider literature sources in this staging (such as (X. Zhang et al., 2024), for narrow-band spectral precursors, (Chaerle et al., 2004, 2007) for thermal microclimatic drift, (Shafay et al., 2025) for economic barriers, and (Malche et al., 2026) for TinyML constraints) is highly deliberate and structurally necessary to map out the specific, evidence-backed bottlenecks for each loop node, ensuring that the TRL assessment remains a comprehensive, standalone synthesis.

Table 3 Technology Readiness Level (TRL) staging of the Pre-Symptomatic Crop Intelligence (PSCI) closed-loop reference architecture, mapping each loop node to its functional role, inputs, current literature staging, key bottlenecks, and targeted maturity level.

PSCI loop node	Primary functional role	Hardware & algorithmic inputs	Current literature staging & key references	Primary engineering & adoption bottlenecks	Targeted TRL
Sensing node	Non-destructive, pre-symptomatic capture of sub-visual physiological stress precursors.	Transmembrane potential difference (EP, mV) (Tran et al., 2019); ChlF kinetics (F_v/F_m , NPQ, R_{fd}) (Moustaka & Moustakas, 2023); (Park et al., 2024); SWIR water-absorption (1400 nm) and red-edge (750 nm) reflectance (X. Zhang et al., 2024; Zhu et al., 2017).	TRL 4 (Lab / Greenhouse Validation): Pre-symptomatic detection of water deficit, nitrogen stress, and viral infections confirmed under controlled environments (Tran et al., 2019; Xu et al., 2026; Zhu et al., 2017).	Extreme sensitivity of non-contact thermal and optical sensors to ambient solar and wind fluctuations (Chaerle et al. 2004, 2007); high CAPEX of research-grade hyperspectral systems (García-Vera et al., 2024; Shafay et al., 2025).	TRL 7 (Field Demonstration)
Inference node	On-device, edge-level processing and temporal onset forecasting of stress trajectories.	1-D time-series electrical potentials (Tran et al., 2019); compressed 2-D spectral frames (SPA-selected) (Zhu et al., 2017); localized meteorological and microclimate forecasts (Elango et al., n.d.).	TRL 4 (Component Validation in Lab): Lightweight CNN backbones (Tiny-LiteNet) and INT8 quantization deployed on single-board computers (RPI 5) (Gookyi et al., 2024; Nyakuri et al., 2025).	Memory and energy constraints of low-cost microcontrollers (Malche et al., 2026); severe lack of dynamic temporal forecasting models to replace static classifiers (Kallenberg et al., 2025).	TRL 6 (Prototype in Operational Environment)
Decision & confidence gating	Uncertainty quantification and post-hoc calibration to filter out background noise and prevent false-alarm cascades.	Softmax probabilities (MSP) (Xiang et al., 2026); predictive entropy (Hasan et al., 2025); temperature scaling (T) (Xiang et al., 2026); Grad-CAM spatial activation mass (Hasan et al., 2025; Xiang et al., 2026).	TRL 3 (Analytical Proof of Concept): Characterization of cross-domain accuracy collapses and the insufficiency of standard post-hoc calibration (selective risk >0.51) under c controlled-to-field transfer (Hasan et al., 2025; Xiang et al., 2026).	Severe domain shift causes classifiers to attend to soil/weed backgrounds rather than central lesions (Xiang et al., 2026); lack of standardized confidence-gating protocols (Xiang et al., 2026).	TRL 6 (Prototype in Operational Environment)
Actuation node	Dynamic, real-time delivery of site-specific variable-rate treatments.	Spatial prescription maps (W. Li et al., 2025); PWM duty cycle and frequency parameters (W. Li et al., 2025); real-time pesticide mixing homogeneity coefficient (γ) (W. Li et al., 2025).	TRL 6 (Operational Ground / UAV Prototype): Pulse Width Modulation (PWM) solenoid valves and inline Venturi mixers achieving a 30% to 50% chemical reduction and over 30% drift reduction (W. Li et	Physical sedimentation of high-viscosity suspension concentrates (SCs) reduces mixing uniformity to 70–75% (W. Li et al., 2025); GNSS-LiDAR temporal errors (>50 ms) cause matching offsets, giving trajectory	TRL 8 (System Commissioned)

			al., 2025).	deviations up to 0.8 m during high-speed field operations (W. Li et al., 2025).	
Feedback & simulation	Continuous updating of the predictive field-state crop twin to simulate “what-if” scenarios and guide sequential interventions.	Leaf Area Index (LAI) and development stage (DVS) observations (Kallenberg et al., 2025); WOFOST and A-scab simulation outputs (Kallenberg et al., 2025); reinforcement learning reward parameters (NUE / yield) (Kallenberg et al., 2025).	TRL 5 (System Validation in Relevant Environment): Interoperable FIWARE context brokers using NGSI-LD and the Subplex algorithm to assimilate in-season observations and run RL-driven decision recommendations in apple and winter wheat fields (Kallenberg et al., 2025).	High computational overhead of running iterative crop simulations (Kallenberg et al., 2025); “sim-to-real” gap where simulated crop twins diverge from heterogeneous field dynamics (Kallenberg et al., 2025).	TRL 7 (Field Demonstration)

Abbreviations: ChlF, chlorophyll fluorescence; EP, electrical potential; NPQ, non-photochemical quenching; SWIR, short-wave infrared; SPA, successive projections algorithm; CNN, convolutional neural network; INT8, 8-bit integer; RPi, Raspberry Pi; MSP, maximum softmax probability; PWM, pulse-width modulation; UAV, unmanned aerial vehicle; SC, suspension concentrate; GNSS, global navigation satellite system; LAI, leaf area index; DVS, development stage; NUE, nitrogen use efficiency; RL, reinforcement learning; TRL, technology readiness level.

Research needs and future directions

The transition of Pre-Symptomatic Crop Intelligence (PSCI) from laboratory-controlled experiments to autonomous field operations requires systematically addressing the technical and economic bottlenecks identified across the cyber-physical loop (Kallenberg et al., 2025). As established in Table 3, while individual sensing and actuation components have achieved high maturity within isolated environments, bridging the gap between localized validation (TRL 4) and full system commission (TRL 8) represents the primary challenge for the precision agriculture community (W. Li et al., 2025). This section outlines five critical research priorities, structured as concrete, fundable problem statements, to guide future development and accelerate the adoption of anticipatory crop protection.

Lead-time-labeled and confidence-reporting benchmarks

Existing agricultural computer vision benchmarks—such as PlantVillage and its derivatives—focus almost exclusively on static, leaf-level disease classification without capturing the temporal dynamics or the physiological precursors of stress development (Shafay et al., 2025). Consequently, models trained on these datasets remain fundamentally reactive, learning to classify conspicuous visual symptoms rather than predicting sub-visual physiological changes (Shafay et al., 2025). Furthermore, these benchmarks do not report the chronological progression of infection or the signal lead time, making it impossible to evaluate a model’s anticipatory capability.

To resolve this bottleneck, we propose a concrete, fundable problem statement: **the establishment of a lead-time-labeled, multi-modal, and confidence-reporting benchmark dataset for tropical and perennial crops**. This proposed project should acquire synchronized, high-frequency measurements—spanning non-contact canopy reflectance (multispectral/hyperspectral), thermal transients, and Photosystem II (PSII) chlorophyll fluorescence kinetics, combined with contact-based stem electrical impedance spectroscopy (EIS) and petiole electrophysiology—across a minimum of three distinct growing seasons. Crucially, each data sample must carry a dual label: (a) a continuous physiological lead-time timestamp (L_t) tracking the duration between the earliest sub-visual signal onset and eventual visual symptom manifestation, and (b) a pathologist-verified pathogen load. Under this standardized benchmark, evaluating deep learning models must mandate the reporting of selective prediction risk, expected calibration error (ECE), and out-of-distribution (OOD) rejection rates, forcing a shift in the literature from uncalibrated top-1 accuracy to calibrated, risk-graded detection confidence (C_d).

Calibrated and uncertainty-aware edge inference

The severe domain-transfer reliability gap documented in the literature highlights that high in-domain laboratory accuracy does not translate into field reliability (Shafay et al., 2025; Xiang et al., 2026). When deep

architectures are evaluated under open-field conditions, they exhibit an overconfident-collapse phenomenon, maintaining high max predicted probabilities (e.g., averaging 79.76%) despite catastrophic drops in actual classification accuracy (dropping to 32.05%) (Xiang et al., 2026). Under the closed-loop PSCI framework, such overconfident false-alarm cascades are economically and agronomically unacceptable because they trigger costly, unnecessary preventative variable-rate spray treatments on healthy foliage, inducing chemical waste and localized phytotoxicity risks (Xiang et al., 2026).

Addressing this calibration gap (Table 3, Node 3) requires the development of lightweight, uncertainty-aware edge inference engines that report reliable, mathematically grounded confidence scores. Rather than relying on computationally heavy deep ensembles or Bayesian neural networks that exceed the memory and power budgets of low-cost microcontrollers, future work must focus on optimizing post-hoc calibration and selective-prediction strategies specifically for edge deployment (Malche et al., 2026).

A key research need is the formulation of on-device uncertainty quantification (UQ) algorithms—such as lightweight temperature scaling calibrated on parent-image-aware local field subsets, or energy-based out-of-distribution scoring—that can execute in under 100 ms on microcontroller-class hardware (Gookyi et al., 2024; Xiang et al., 2026).

This must be paired with selective prediction policies that allow the model to abstain from predicting ($g_s(x) = 0$) when the confidence score falls below a dynamically updated threshold, routing the flagged subplots to autonomous gantry-level high-frequency re-evaluation rather than committing to an overconfident, uncalibrated guess (Xiang et al., 2026).

Distributed learning and geographic generalization

Agricultural AI models frequently experience performance degradation when transferred across different regions, soil types, and climate zones due to the strong environmental variability and spatiotemporal non-stationarity of field data (Shafay et al., 2025). Standard supervised models trained on centralized, localized datasets are highly sensitive to these domain shifts, and forced feature alignment often disrupts task-relevant discriminative structures (Xiang et al., 2026). This represents a severe barrier for smallholders in developing regions who cannot afford high-capacity central servers or continuous mobile data costs to upload images to external cloud platforms (Ibegbu & Korsah, 2024).

To overcome this geographic barrier, future research must exploit distributed learning architectures—specifically federated learning (FL)—to train robust, geographically generalized models without requiring centralized data consolidation (Z. X. Yang et al., 2025). We define a second fundable problem statement: **the design of an edge-collaborative federated learning framework for crop intelligence**. This framework should enable independent, low-cost smallholder edge nodes (such as ESP32-S3 microcontrollers or portable smartphones) to perform local model training on local, heterogeneous stress signatures, transmitting only lightweight weight updates or high-level semantic features rather than raw, high-dimensional data (Ibegbu & Korsah, 2024; Z. X. Yang et al., 2025). Research must target the development of multi-view calibration networks and dynamic adaptive fusion weights to handle extreme data heterogeneity across distributed nodes, ensuring that the aggregated global model achieves robust cross-geographic generalization while strictly safeguarding smallholder data privacy and minimizing communication overhead (Z. X. Yang et al., 2025).

Temporal onset forecasting models

A major conceptual gap in the current literature is the reliance on static categorical classifiers that evaluate a crop's health state at a single, isolated instant (Shafay et al., 2025). Under natural field conditions, crop stress is a continuous, dynamic physiological trajectory governed by weather variables and microclimatic predisposing factors (Elango & Nelson, 2020). Standard classifiers fail to capture these temporal dependencies, leading to unstable predictions under rapid environmental fluctuations. While historical time-series forecasting models—such as the ARIMAX(0,2,1) model fitted with maximum temperature as a covariate to predict weekly Rugose Spirling Whitefly (RSW) nymph surges—have established the viability of incidence forecasting, they remain linear and struggle to model non-linear physiological-meteorological interactions (Elango et al., n.d.).

Future directions must focus on extending these linear baselines into deep temporal sequence-forecasting architectures deployed directly at the edge. Research should investigate the integration of multi-head self-attention

networks, Temporal Fusion Transformers (TFTs), and bidirectional recurrent architectures (such as Bi-LSTM or Bi-GRU) to process continuous, multi-channel 1-D time-series data from microclimate sensors and plant electrophysiological arrays in parallel (Z. X. Yang et al., 2025). These models must be designed to output multi-horizon sequential predictions across multiple future time steps, estimating the exact "time-to-peak" of the stress trajectory under variable weather forecasts (Elango et al., n.d.). By transforming static classification into continuous temporal forecasting, the decision support system can schedule preventative variable-rate actuation precisely within the pre-symptomatic window, resolving the actuation-latency bottlenecks highlighted in Table 3 (W. Li et al., 2025).

Economic and operational evaluations for perennials

While the pre-symptomatic capture of crop stress has achieved high laboratory accuracy, the long-term economic viability and multi-seasonal operational return on investment (ROI) of closed-loop PSCI architectures remain largely unquantified in the literature (Shafay et al., 2025). The capital expenditure (CAPEX) of advanced sensing systems represents an acute adoption barrier, with research-grade hyperspectral cameras costing between \$20,000 and \$50,000 compared to low-cost RGB action cameras or custom multi-sensor IoT nodes (costing under \$25 to \$100) (Legendre et al., 2021; Shafay et al., 2025). Because perennial orchards and vineyards represent significant, multi-year capital investments where unchecked stress vectors can permanently damage the long-term woody architecture of the crop, a rigorous quantification of the economic feedback loop is required to justify these high early-stage CAPEX costs (Nackley et al., 2021; Shafay et al., 2025).

To establish this economic foundation, we propose a third fundable problem statement: **a multi-seasonal, double-blind randomized controlled field trial in commercial perennial systems to evaluate the economic ROI of pre-symptomatic interventions.** This study should establish side-by-side experimental blocks comparing: (1) a closed-loop PSCI variable-rate spray and irrigation treatment (triggered pre-symptomatically via fused electrophysiological, thermal, and ChlF sensors), (2) a conventional constant-rate spray treatment (triggered reactively at the visual symptom stage), and (3) an untreated control. The primary outcome metrics must include: (a) total chemical input volume and pesticide drift reduction, (b) multi-seasonal yield and quality preservation, (c) long-term crop vigor and wood-architecture health, and (d) the amortized payback period of the sensory-actuation hardware. By quantifying these multi-year, field-scale economics, this research will supply the empirical ROI evidence required to drive commercial adoption and establish pre-symptomatic precision plant protection as a validated Drift Reduction Technology (DRT) globally (Nackley et al., 2021).

Conclusion

The transition from reactive, visual-stage crop disease management to proactive, anticipatory plant protection represents a critical frontier in modern precision agriculture. This systematic review presented Pre-Symptomatic Crop Intelligence (PSCI), a formal cyber-physical closed-loop reference architecture that reframes sub-visual stress detection from an isolated classification challenge into an integrated, decision-oriented site-specific intervention loop. At the core of this paradigm sits the coupled system-value thesis: the realized agronomic and economic utility of any pre-symptomatic framework is strictly bounded by its weakest cyber-physical link. This relationship is mathematically governed by physiological lead time (L_t), calibrated detection confidence (C_d), and actuation actionability (A_c), such that if any of these variables approaches zero, the overall value of the system collapses ($V \rightarrow 0$). By prioritizing loop symmetry over the optimization of isolated sensing or machine learning elements, the PSCI framework protects growers from both costly false-alarm cascades and late, ineffective interventions.

This review established four key contributions to synthesize and guide this emerging domain. First, the closed-loop cyber-physical reference architecture was structured across five interconnected nodes, systematically mapping previous literature streams from raw physiological precursors to predictive digital-twin updates. Second, a risk-graded decision-space framework was established, partitioning the $L_t \times C_d$ plane into actionable operational regimes (Act Now, Monitor, Defer) to guide site-specific treatments based on quantitative uncertainty gates. Third, a predictive, constrained digital-twin feedback loop was integrated to assimilate post-actuation physical field states, continuously updating and refining subsequent onset forecasts. Fourth, the generalizability of the closed loop was demonstrated across three recurring crop stress cases, using the principal biotic pest, the Rugose Spiraling Whitefly

(RSW) on coconut, alongside abiotic water and nitrogen deficits, to demonstrate the biological breadth of the PSCI architecture.

Ultimately, the commercial deployment and scalability of pre-symptomatic crop intelligence depend on its accessibility in challenging, resource-constrained environments. Translating high-throughput laboratory and greenhouse accuracies into robust open-field performance requires addressing the specific hardware, environmental, and adoption constraints of tropical smallholder and perennial cropping systems. By focusing research efforts on the critical priorities identified in this work—namely, lead-time-labeled benchmarks, uncertainty-aware edge inference, field-robust fusion, and multi-seasonal economic evaluations—the precision agriculture community can move beyond visual-stage diagnostics. The realization of this anticipatory-agriculture vision will provide growers with a highly calibrated, low-cost, and environmentally sustainable system of preventative plant protection, safeguarding global crop yields and agricultural ecosystems before symptoms ever appear.

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